

Synergistic Effect of Blended Components in Nonaqueous Electrolytes for Lithium Ion Batteries

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Abstract Application of different electrolyte components as blends in nonaqueous electrolyte formulations represents a viable approach towards improving the overall performance and reliability of a lithium ion battery cell. By combining the advantages of different electrolyte constituents, cell chemistry can be optimized and tailored for a specific purpose. In this paper, the current progress on possibilities, advantages, as well as limitations of blended nonaqueous electrolyte formulations, including solvent, salt and additive blends is reviewed and discussed. Emphasis is set on the physicochemical, electrochemical, and safety aspects. In addition, the aim of this review is to provide perspective and possible strategy for further and future development of blended nonaqueous electrolytes with long life, high energy density, high power, and adequate safety at competitive manufacturing costs. The provided overview and perspective on blended nonaqueous electrolyte formulations should encourage researchers to proceed with further and deeper investigations in this promising field of advanced batteries.

Keywords Blends · Synergistic effect · Nonaqueous electrolytes · Electrolyte solvent · Conducting salt · Additive

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1 Introduction

Among a wide variety of different storage technologies for electric energy, electrochemical energy storage in batteries was recognized as the most promising for portable electronics, stationary and (auto-)motive applications [1]. Thanks to the high energy efficiency [2, 3] and outstanding scalability, batteries are key elements in all energy scenarios nowadays. Twenty-five years after the commercialization by Sony Corporation, lithium ion batteries (LIBs) have significantly improved in performance, reliability, as well as safety, thus representing an accustomed part of our everyday life [4].

The key performance requirements of a battery cell for different applications might look very similar at first sight; however, each application has different performance requirement priorities. The cycle life of a rechargeable battery cell is strongly influenced by the long-term reversibility of the redox reactions represented by the cell chemistry. The crucial factor maintaining this reversibility is related to chemical as well as electrochemical stability of the used electrolyte formulation. The nonaqueous electrolyte is the key component for the development of novel and improved cell chemistries, custom-tailored for the specific application. One of the crucial points for next-generation lithium batteries is associated with the development of high-performance electrolyte components, namely solvents, conducting salts, and additives in replacing the current state-of-the-art (SOTA) organic carbonate solvent-based [5] electrolyte formulations with lithium hexafluorophosphate (LiPF_6) as conducting salt [6]. These novel and improved electrolyte formulations should assure high ionic conductivity as well as high thermal and electrochemical stability, long-term stability vs. anode and cathode over a wide state of charge (SoC) range and lead to formation of effective solid electrolyte interphases (SEI) [7] and cathode electrolyte interphases (CEI). At the same time, the aforementioned (ideal) electrolyte components should be non-flammable, nontoxic and environmentally friendly (Fig. 1).

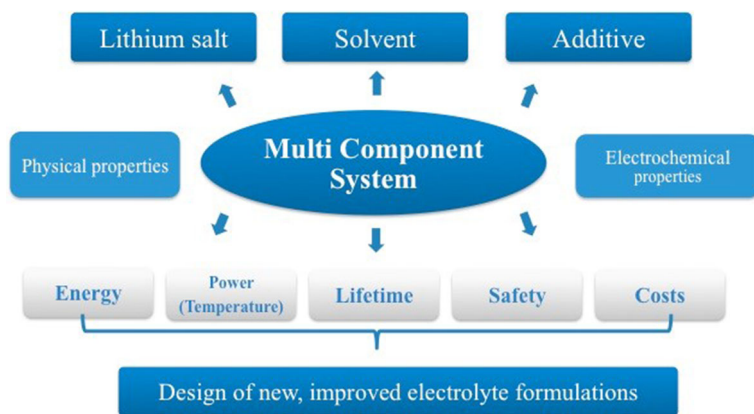


Fig. 1 Nonaqueous electrolyte as core and key component of a lithium ion battery (LIB) cell

One way in improving the SOTA nonaqueous electrolyte formulations is to synthesize and thoroughly characterize new electrolyte components to determine their physicochemical properties and electrochemical behavior. With this in line, the focus of many research activities is set on development of new electrolyte formulations through new solvents, new salts, or on modifying conventional systems with adequate electrolyte additives. However, functionalities with the beneficial impact on the cell performance are usually obtained at the cost of other characteristics. For this reason, blends containing at least two chemical substances with different molecular structures of the same electrolyte component (solvent, salt, and additive) can be used.

The utilization of blended component electrolytes represents an alternative approach towards maximizing the performance of lithium ion cells by combining the advantages of each constituent. The synergy between two or more different solvents, conductive salts or various additives, as a blend in an electrolyte formulation, has the potential to significantly contribute to the desired advancements of the lithium ion technology. In this regard, blends of electrolyte components selected in adequate amounts, perform complementary functions to each other. By definition, a synergistic effect arises from the interaction between two or more substances, agents, entities or factors that produces an effect greater than the sum of the individual ones. Taking the term synergy as a word to its roots, “syn” has the meaning of together whereas “ergon” means work. A strong synergistic effect of properly selected components may have a positive outcome on the overall performance, as blending of ‘right’ electrolyte components enhances the physicochemical, thermal, as well as electrochemical properties of all of them. In this regard, a mixture of high dielectric solvents (HDS—solvate ions and favor salt dissociation, but highly viscous) and a low viscosity (LVS—facilitate ion transport, low permittivity) solvents used in common SOTA electrolytes is a well-known example of the synergistic effect influencing the ionic conductivity of the electrolyte. This important measure of the conductivity through ionic charge carriers, which determines internal resistance and electrochemical behavior of the cell, is dependent on the nature as well as on the ratio of the solvents in the electrolyte system. In order to reach the maximum ionic conductivity, an optimum HDS + LVS composition has to be identified (Fig. 2a). Beside properly selected solvent mixture in an electrolyte formulation, a Li salt (LiX) concentration plays an important role in maximizing the ionic conductivity. As depicted in Fig. 2b, at low salt concentrations, which is in solvent excess, the number of ions as charge carriers will be low. However, it will consequently increase with the increase of salt concentration until reaching a maximum at certain point. The location of this conductivity maximum is dependent on the dielectric constant of the solvent(s) used in the electrolyte as well as the temperature. In solvent-mixed electrolytes containing solvent with high dielectric constant or at higher temperatures at which electrolyte viscosity is decreased, the conductivity maximum will be shifted to higher LiX concentrations. After passing through a conductivity maximum, the number of free charge carriers as well as the ionic mobility will decrease with further increase of the LiX concentration, due to the intense ion pair formation and

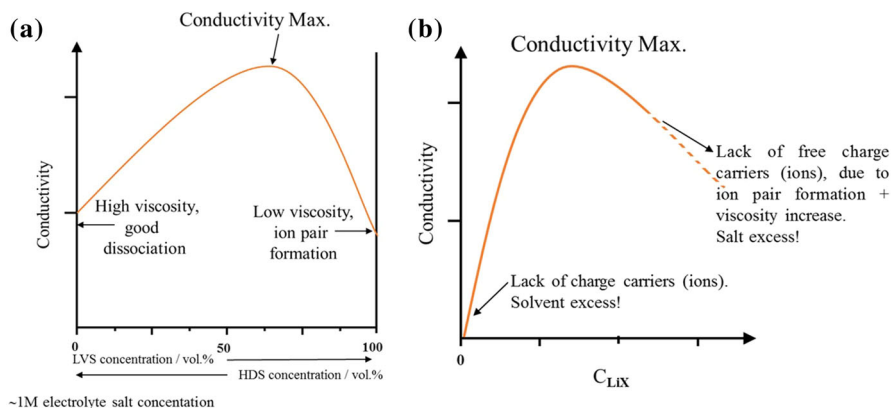


Fig. 2 Ionic conductivity dependency on concentration of **a** HDS and LVS in mixed electrolyte solution and **b** lithium salt [8]

higher viscosity of the electrolyte formulation. At very high LiX concentrations, the salt may even precipitate.

This survey aims to highlight the opportunities and perspectives of nonaqueous electrolyte formulations containing different classes of solvents, salts and additives as blends in LIB cells, without neglecting their limitations.

2 Solvent Blends in Nonaqueous Electrolytes

Despite the fact that the organic carbonate-based nonaqueous electrolyte is the most suitable choice for graphite-based LIBs due to formation of an effective SEI layer on the graphite surface, the safety problems at only moderately elevated temperatures ($>40^\circ\text{C}$) are a matter of concern for industry as well as for consumers [9, 10]. Furthermore, the electrochemical stability of the SOTA organic carbonate solvent-based electrolyte limits the high-voltage operation, required for high-energy LIBs in electric vehicles to extend energy density and driving range. Thus, there is an unmet need for development of alternative electrolyte formulations with higher electrochemical stabilities and improved safety properties [11]. A possible solution for substituting carbonate-based solvents refers to selection of other classes of compounds. Among them, nitriles and ionic liquids (ILs), showed significant potential due to their thermal stability over a broad temperature range, wide electrochemical stability window (ESW) and in case of the ILs and many (but not all) nitriles, negligible vapor pressure [12–17]. In recent years, mono- and dinitriles received considerable attention as promising solvents for LIB electrolytes [18, 19]. Due to the high thermal and good anodic stabilities, especially in domain of high-voltage operation, special focus was set on dinitriles, such as glutaronitrile (GLN), adiponitrile (ADN), or sebaconitrile (SEN). The importance of the solvent flash point was emphasized by Wrodnigg et al. [20] as early as 1999. Abu-Lebdeh et al. [15, 16] and Nagahama et al. [17] proposed GLN, ADN, and SEN as possible

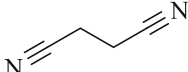
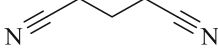
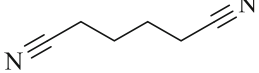
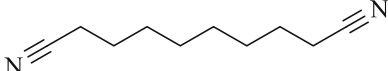
solvents for electrolytes in LIBs. Among them, GLN and ADN were recognized as the most attractive candidates owing to the highest values of boiling and flash points, highest dielectric constants and lowest viscosities. However, nitriles as in single solvent electrolytes resulted in a low ionic conductivity (3.6 mS cm^{-1} for GLN and 1.8 mS cm^{-1} for ADN at 25°C) and are incompatible with graphite, as shown by Li et al. [21]. Abu-Lebdeh et al. [15] revealed that ADN as single solvent, either with lithium bis(trifluoromethylsulfonyl)imide (LiTFSI) alone or together with the SEI-promoting additive, lithium bis(oxalato)borate (LiBOB), led to only limited capacities in graphite/LiCoO₂ (LCO) full cells. Moreover, the melting points were too high for battery applications. The drawbacks of using nitriles as single solvents in LIB cells clearly indicated that for better battery performance, blending is required. Besides nitriles, systematic research on the application of ILs as electrolyte solvents in energy storage devices was increased in recent years. The most popular cations of room temperature ionic liquids (RTILs) are quaternary ammonium-based such as tetraalkylammonium, aromatic cyclic amines (pyridinium, imidazolium) and saturated cyclic amines (pyrrolidinium, piperidinium), whereas the most used counter anions refer to non-coordinating anions such as [B(CN)₄], bis(trifluoromethylsulfonyl)imide (TFSI) and bis(fluorosulfonyl)imide (FSI)—very large anions with small charge to radius ratios. This leads to weaker electrostatic bonding between cation and anion as well as to lower melting points. Low lithium ion mobility, which is a direct consequence of the higher viscosity of RTILs, is in fact, one of the main obstacles in performance of LIBs using RTIL-based electrolytes [22, 23].

2.1 Nitrile-Based and Ionic Liquid-Based Solvent Blends

2.1.1 Aim of Solvent Blending

As listed in Table 1, the melting points of the dinitriles $\text{CN}(\text{CH}_2)_n\text{CN}$ with $n = 2, 3, 4, 8$ follow no regularity and the values for succinonitrile (SCN), ADN and SEN are

Table 1 Melting and flash point values of selected dinitriles $\text{CN}(\text{CH}_2)_n\text{CN}$ with $n = 2, 3, 4, 8$

Name	$\text{CN}(\text{CH}_2)_n\text{CN}$ $n =$	Structural formula	Melting point ($^\circ\text{C}$)	Flash point ($^\circ\text{C}$)
Succinonitrile	2		50–54 [23]	113 [23]
Glutaronitrile	3		–29 [22]	113 [22]
Adiponitrile	4		1–3 [21]	163 [21]
Sebaconitrile	8		7–8 [24]	>113 [24]

in the positive range, limiting their performance at low temperatures. For this reason, blending nitriles with other solvents enables the utilization of nitriles for LIB application.

Ethylene carbonate (EC), a cyclic carbonate which is present in conventional carbonate-based electrolytes in LIBs, has a high melting point (35–38 °C [25]) and can therefore not be used as single solvent at room temperature. Blending of EC and nitrile led to a binary eutectic blend with a lower melting point [≈ -60 °C for EC/ acetonitrile (ACN)] [26]. It is well known that the linear carbonates, e.g., dimethyl carbonate (DMC) and EC also form an eutectic blend with a lower melting point [27]. The addition of a nitrile to the carbonate blend generates a ternary eutectic blend. Based on the EC/DMC/ACN blends, Cho et al. showed that by using ≈ 60 vol% ACN and ≈ 40 vol% EC or EC/DMC (1:1 v/v), the eutectic point can be attained [26]. Moreover, the eutectic point at ≈ -60 °C could be reached with a binary or a ternary blend. The phase diagram, shown in Fig. 3c, was generated by plotting the temperatures of the endothermic peaks versus volume fraction of EC/ DMC obtained from the differential scanning calorimetry (DSC) thermograms (Fig. 3a, b).

A single solvent nitrile system most often fails to fulfill major demands for the commercial applications. An exception was constituted by a single-solvent electrolyte based on cyanoesters which combined adequate electrochemical performance with improved safety [28]. Nevertheless, blending of several solvents brings multifunctional solution to electrolyte formulation. Nitriles have high boiling and flash points, whereas conventional carbonate-based solvents suffer from relatively lower boiling and flash points; thus their high flammability is an issue, which endangers safe operation of a LIB. For this reason, blending represents a promising way to improve the safety of nonaqueous electrolyte formulations. Additionally, a solvent blend containing nitrile and carbonate has a reduced viscosity due to the low viscosities of nitriles. In such solvent blend, carbonates take

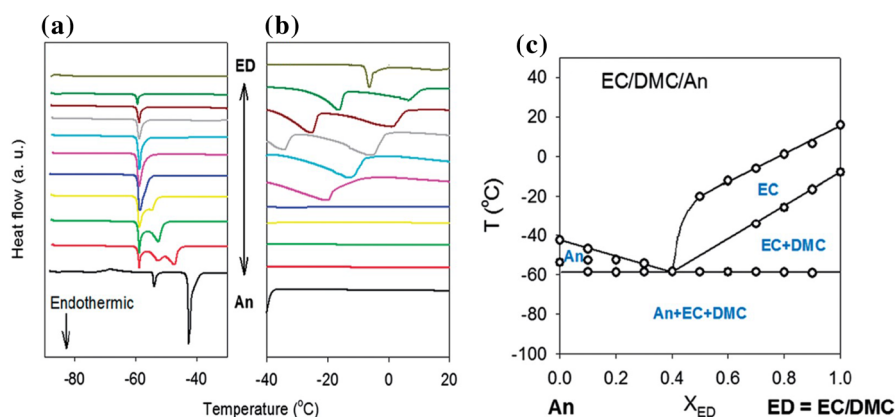
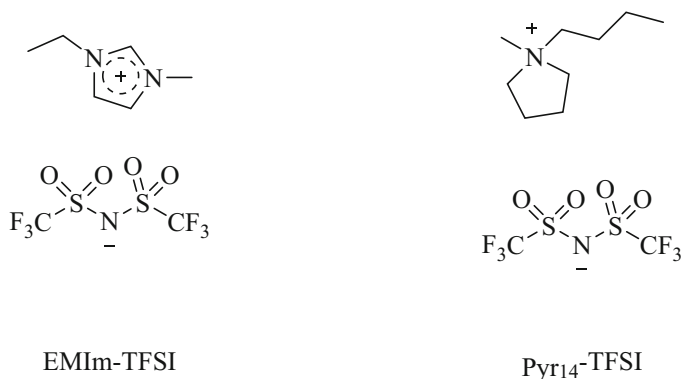


Fig. 3 Thermograms for ACN/EC/DMC solvent blends measured by DSC. Endothermic heat flow is downward (**a**, **b**). Heat flow was normalized in **b** to clarify the peaks. **c** Resulting phase diagram, ED = EC/DMC (1:1 v/v). X_{ED} = volume fraction of ED. Reprinted from Ref. [26]. Copyright (2014) with permission of Royal Society of Chemistry



Scheme 1 Molecular structures of 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide (EMIm-TFSI) and *N*-butyl-*N*-methylpyrrolidinium bis(trifluoromethylsulfonyl)imide (Pyr₁₄-TFSI)

over the task related to the formation of an effective SEI and make the electrolyte more conductive. This leads to an assumption that nitriles and carbonates complement each other, thus resulting in a better electrochemical performance.

To clarify the effect of RTILs on the safety of LIBs, Guerfi et al. investigated the flammability of RTILs and carbonate solvent blends [29]. They prepared several electrolyte blends of the IL 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide (EMIm-TFSI, Scheme 1) and 1 M LiPF₆ in EC/diethyl carbonate (DEC) (1:1 by wt.) containing 2% vinylene carbonate (VC), with different concentrations of RTIL. The electrolyte with 100% organic carbonate started to burn in the first second of ignition, whereas in pure RTIL, no combustion occurred even after 20 s of flame exposure. The flame exposure time was increased with addition of EMIm-TFSI to the carbonate-based electrolyte. The flammability of the blended electrolyte was prevented by addition of 40% or more EMIm-TFSI. An optimum composition for better battery performance (low viscosity and high conductivity) and higher safety (no flammability) was proposed with 40–60% EMIm-TFSI.

Arbizzani et al. carried out flammability studies in blends composed of 1 M LiPF₆ in EC/DMC (1:1 by wt.) as carbonate-based electrolyte and various concentrations of the *N*-butyl-*N*-methylpyrrolidinium bis(trifluoromethylsulfonyl)imide (Pyr₁₄-TFSI, Scheme 1) with a slightly different setup (Fig. 2) [30]. A glass fiber mat (Whatman, GF/F) was soaked with the investigated electrolyte blends and exposed to the flame placed at a distance of 123 mm from the top of the burner tube so that the ignition was only due to the temperature increase. The flame was turned on for 5 s.

Each sample was measured six times and the average value ($\approx 10\%$ error) of the self-extinguishing time (SET) was used. Table 2 lists the reproducibility data of the ignition occurrence as an indication of the flame inhibition effect of different electrolyte blends. At least 30 wt% RTIL should be added to a carbonate-based electrolyte in order to observe the flame-inhibition effect. With addition of more than 50 wt%, the tendency to ignite was significantly reduced. Considering the

Table 2 Ignition occurrence and the mean values of the SET of several blends of 1 M LiPF₆ in EC/DMC (1:1 by wt.) and Pyr₁₄-TFSI. Reprinted from Ref. [30]. Copyright (2011) with permission of Elsevier

Pyr ₁₄ -TFSI (wt%)	Ignition occurrence	SET (s g ⁻¹)
0	6/6	55
10	6/6	67
30	4/6	109
50	2/6	125
70	2/6	127
100	0/6	–

major safety problems of carbonate-based electrolytes as well as certain unfavorable physicochemical properties (melting point and viscosity) of single solvent electrolytes containing nitriles and RTILs, blending of carbonate solvents with nitriles or RTILs could be a solution to overcome afore-described drawbacks.

2.2 Effect of Blending on Physicochemical Features

In order to formulate an electrolyte with a higher flash point and at least steady electrochemical performance, Isken et al. [31] mixed EC with ADN in 1:1 (by wt.) ratio. The flash point of this blend was found to be more than 110 °C higher than blends of EC with a linear carbonate like DMC or DEC. Furthermore, the vapor pressure of such blend is much lower and has a positive effect on the safety of cells.

Duncan et al. [32] revealed viscosity values of 1 M LiTFSI in pure dinitrile to be much higher than in the binary solvents EC/dinitrile (1:1 v/v) at room temperature. Due to the fact that LiBOB as co-salt leads to very good cycling performance, the authors used the ternary solvent blend containing EC/DMC/dinitrile (1:1:2 v/v/v) in combination with a salt blend [1 M lithium tetrafluoroborate (LiBF₄) + 0.1 M LiBOB] to obtain the lowest viscosities (Fig. 4).

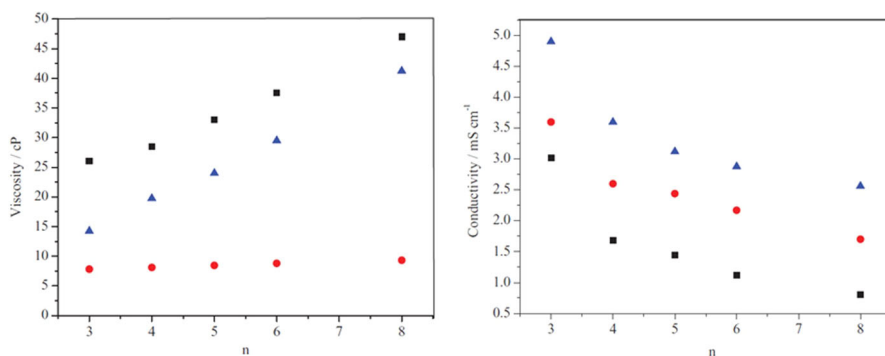


Fig. 4 Viscosity (left) and conductivity (right) values of 1 M LiTFSI in pure dinitrile (black filled square), 1 M LiTFSI in EC/dinitrile (1:1 v/v) (blue filled triangle) and 1 M LiBF₄, 0.1 M LiBOB in EC/DMC/dinitrile (1:1:2 v/v/v) (red filled square) at room temperature. Dinitriles with different chain lengths: NC-(CH₂)_n-CN with $n = 3$ –8. Reprinted from Ref. [32]. Copyright (2013) with permission of ECS

It became apparent that a longer alkyl chain of the dinitrile gives rise to a higher viscosity of the electrolyte. The longer alkyl chain is related to a bigger molecular size, which leads to decreased mobility and conductivity. The electrolytes containing 1 M LiTFSI in a single dinitrile $(\text{NC}-(\text{CH}_2)_n-\text{CN})$ with $n = 3-8$ showed the lowest conductivities whereas the investigated EC/dinitrile blends with 1 M LiTFSI revealed the highest conductivities. Furthermore, the conductivities of the electrolytes with binary solvent blends were higher than those with ternary solvent blends (1 M LiBF_4 + 0.1 M LiBOB in EC/DMC/dinitrile), although the last-named had the lowest viscosities. The reason behind this is associated with the fact that the ternary blend has a lower dielectric constant than the binary blend; and LiBF_4 , as a different conducting salt, has a lower dissociation constant. As a result of weak salt dissociation, the ionic conductivity of the electrolyte significantly decreased.

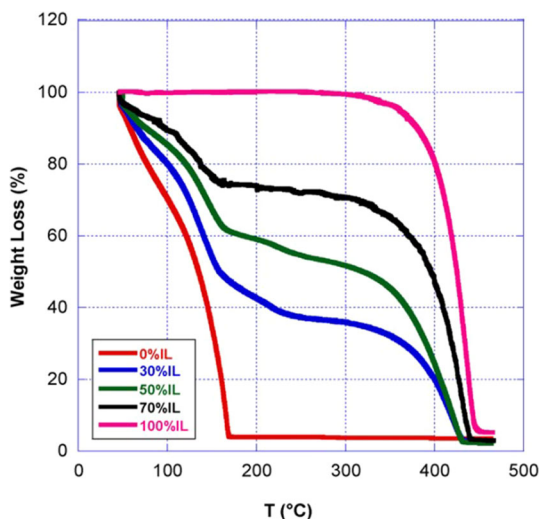
Abu-Lebdeh et al. [16] showed that addition of EC to GLN led to an increase in conductivity and to a decrease in viscosity. The temperature dependence as well as the dependence on the salt concentration of LiTFSI was found to be similar for both electrolytes. Increase in temperature resulted in higher conductivity values. At room temperature, the highest conductivity was reached with a 1 M solution of LiTFSI in EC/GLN (1:1 v/v) amounting to 4.9 mS cm^{-1} . Higher concentrations caused lower conductivities, most probably due to ion pairing or increased viscosity.

Isken et al. [31] revealed that a blend of 0.9 M LiBF_4 in EC/ADN (1:1 by wt.) showed better conductivities at room temperature compared to the 1 M LiBF_4 in EC/DEC (3:7 by wt.) electrolyte.

MacFarlane et al. [33] introduced pyrrolidinium-saturated cations with higher cathodic stability on the graphite electrode as well as better interfacial behavior on lithium metal anodes. However, compared to imidazolium-based ILs, conductivity values were lower and viscosities higher. IL blends of electrolytes with carbonate-based solvents can tune the physicochemical and electrochemical properties of different types of RTILs to achieve better performance in the cell. The carbonate solvents could improve the SEI formation capability of the blended electrolytes and in the same time increase the conductivity. Many groups investigated the improvements in conductivity of solvent blended electrolytes and the obtained results pointed out the positive effect of carbonate solvents on reducing the viscosities and improving the conductivities [29, 34].

Thermal stability of the nonaqueous electrolytes is the key to better performance of the battery cell as internal heat is always produced during charge and discharge. As mentioned above, SOTA electrolytes suffer from low thermal stability of LiPF_6 and organic carbonate-based solvents. The thermal decomposition of organic electrolytes as well as blended electrolytes with RTILs was investigated by many groups [29, 30, 35–37]. The general conclusion of these investigations was that the addition of RTILs to the carbonate-based electrolytes have positive effects on the thermal stability of electrolyte blends, but the presence of the RTIL cannot stop the thermal decomposition of LiPF_6 and the carbonate molecules. Guerfi et al. investigated the thermal stability of the pure solvent and solvent blended electrolytes by means of thermogravimetric analysis (TGA) as depicted in Fig. 5 [29].

Fig. 5 Thermogravimetric analysis of 1 M LiPF₆ in EC/DEC and 2% VC with different EMIm-TFSI ratios. Reprinted from Ref. [29]. Copyright (2010) with permission of Elsevier



The pure RTIL showed less than 3% weight loss up to 350 °C. In contrast, 30% weight loss was observed in case of the carbonate-based electrolyte using LiPF₆ as conducting salt, when the temperature was raised up to 100 °C and furthermore about 80% loss at 160 °C. Although the addition of RTILs enhanced the thermal stability of the solvent blended electrolytes, the thermal stability in the temperature range from 25 to 100 °C was not improved.

Arbizzani et al. investigated the thermal decomposition of blended electrolytes under N₂ and N₂/O₂ atmosphere, the latter to simulate air conditions (Fig. 6) [30]. The influence of the atmosphere was found to be negligible. Furthermore, it was shown that DMC and EC evaporated sequentially based on their boiling points (89

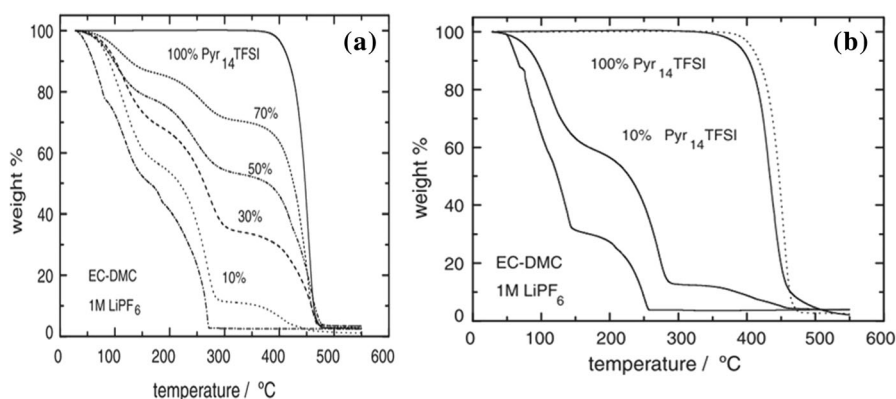
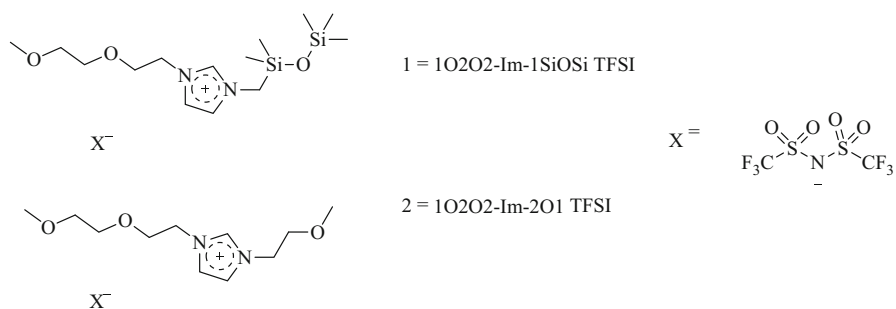


Fig. 6 TGA curves of **a** 1 M LiPF₆ in EC/DMC (1:1 by wt.) and Pyr₁₄-TFSI solutions with different wt% of Pyr₁₄-TFSI under N₂ atmosphere and **b** 1 M LiPF₆ in EC/DMC (1:1 by wt.), Pyr₁₄-TFSI and a solution with 10% Pyr₁₄-TFSI under N₂/O₂; for comparison the curve of Pyr₁₄-TFSI under N₂ atmosphere (dotted line) is also reported. Reprinted from Ref. [30]. Copyright (2011) with permission of Elsevier



Scheme 2 Ether–ether and ether–siloxane ILs [41]

and 260 °C) and the thermal decomposition of LiPF_6 in electrolyte solution starting at temperatures below 107 °C [38–40].

Chavan et al. investigated thermal decomposition of two newly synthesized RTILs (Scheme 2) and their blends with propylene carbonate (PC, 55 mol%) in electrolyte containing 1 M LiTFSI as conductive salt [41].

Both, pure ILs and ILs in combination with LiTFSI lost 90% of weight in the temperature range from 350 to 530 °C. The IL + PC solvent blended electrolytes with LiTFSI showed two major weight loss steps at ≈ 246 °C (15% weight loss), which could be due to decomposition of the PC component, and at 350 °C with 75% weight loss originating from IL decomposition. The addition of LiTFSI to blends of PC and investigated ILs led to the improvement of thermal stability of ether–ether IL + PC blends (from 110 to 150 °C, Fig. 7a) and for ether–siloxane IL + PC blends (from 100 to 130 °C, Fig. 7b). Similar improvements resulted from addition of LiTFSI to ILs (309–355 °C for ether–ether IL and 307–329 °C for ether–siloxane IL). Chavan et al. additionally showed that an ether–ether substituent IL has a higher thermal stability than an ether–siloxane IL under similar conditions.

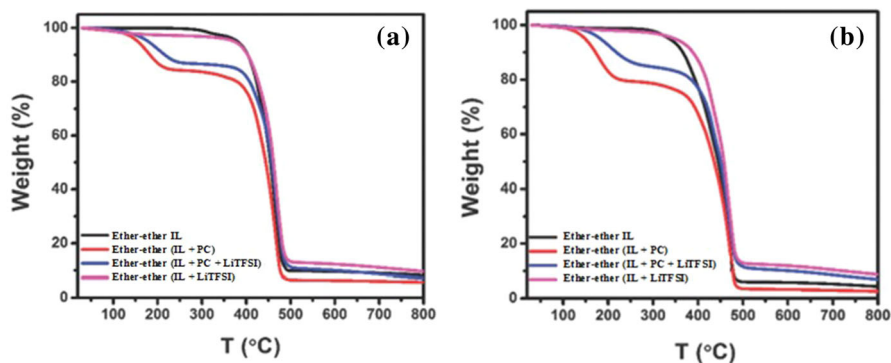


Fig. 7 TGA analysis of **a** ether–ether IL; **b** ether–siloxane IL. Redrawn from Ref. [41]. Copyright (2016) with permission of Royal Society of Chemistry

2.3 Effect of Blending on Electrochemical Performance

2.3.1 Cycling Performance and Electrochemical Stability

Aliphatic nitriles are known for their incompatibility with the graphite anode, as they are not able to form an effective SEI. Abu-Lebdeh et al. suggested that graphite can only be stabilized vs. nitriles by using a blend with SEI forming components (solvents and salts) and proposed to mix ADN and EC together with 1 M LiTFSI + 0.1 M LiBOB salt blend, as the solvent blend only is not sufficient to insure good electrochemical performance [15]. LiBOB is known as a very strong SEI-promoting electrolyte component [42]. ADN and EC decompose at 0.4 and 0.7 V vs. Li/Li⁺ respectively, whereas BOB[−] reduction occurs at more positive potentials (1.5 V vs. Li/Li⁺), thus stabilizing the surface of the graphitic structure. Xu et al. reported that the presence of EC and LiBOB influences the formation of an effective SEI [42]. With this in line, the authors investigated electrochemical performance of EC/ADN blends (1:1 v/v) compared to pure ADN in the presence of 1 M LiTFSI as conducting salt. Cycling results obtained in graphite/LCO full cells at a C-rate of C/12 showed much higher initial capacity for the EC/ADN blends (1:1 v/v). Using 0.1 M LiBOB as co-salt reduced the capacity fading. The blend of 1 M LiTFSI, 0.1 M LiBOB in EC/ADN (1:1 v/v) showed the best cycling performance with an initial specific discharge capacity of 108 mAh/g and a capacity retention of 90% after 50 cycles (Fig. 8). The reductive decomposition of ADN on graphite was prevented by the synergistic effect between EC, BOB[−] and TFSI[−] anions resulting in an effective SEI formation [15].

Isken et al. revealed that an EC/ADN (1:1 by wt.) solvent blend containing 0.9 M LiBF₄ shows good cycling performance without considerable capacity fading [31].

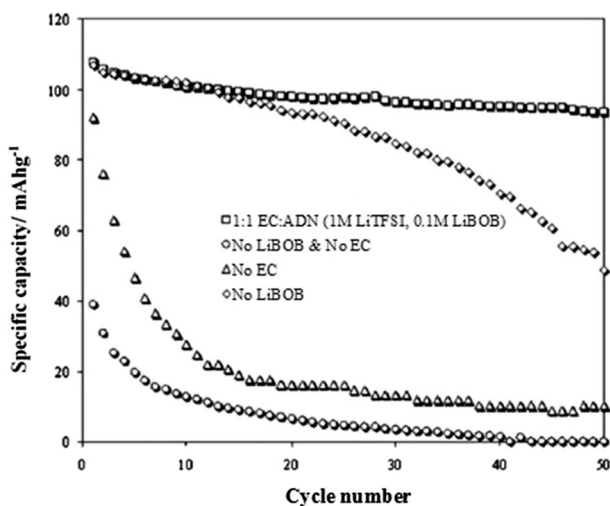


Fig. 8 Effect of electrolyte components on cycling performance of ADN-based electrolytes at C/12 in graphite/LCO full cells. Redrawn from Ref. [15]. Copyright (2009) with permission of Elsevier

Graphite/Li half-cells showed almost the theoretical capacity, even at high C-rate (5 C). After 50 cycles, the capacity was still more than 98% of the initial value. In case of LiNiMnCoO₂ (NMC)/Li half-cells, a capacity retention of 97% was obtained after 50 cycles. The anodic stability of the EC/ADN blend (6.4 V vs. Li/Li⁺) was 0.3 V higher than that of the standard electrolyte 1 M LiPF₆ in EC/DEC (3:7 by wt.) (6.1 V vs. Li/Li⁺), which better enables the use of high-voltage electrodes. The standard electrolyte showed ≈ 0.2 V higher cathodic stability compared to the dinitrile blend. However, this might be attributed to an additional reduction process that gives rise to passivation of the platinum (Pt) working electrode. The addition of a third solvent, a linear carbonate, such as DMC or ethyl methyl carbonate (EMC), to the binary blend could also have a significant impact on the electrochemical performance of the electrolyte. Duncan et al. disclosed that LiNi_{0.5}Mn_{1.5}O₄ (LMNO)/Li half-cells operate only when DMC as third solvent, LiBF₄ as salt and LiBOB as co-salt are used [32]. Furthermore, they indicated that higher capacities could be reached with shorter alkane dinitriles such as ADN than with longer alkane dinitriles such as SEN. Nevertheless, all investigated electrolyte blends showed similar capacity fading.

Nagahama et al. [17] investigated ternary blends consisting of 1 M LiBF₄ in EC/DMC/SEN (1:1:2 v/v/v). They showed that the ternary blend has unique features such as a wide electrochemical stability window of ≈ 6 V (0.5–6.5 V vs. Li/Li⁺) and a good electrochemical compatibility with Li₄Ti₅O₁₂ (LTO), LiMn₂O₄ (LMO), LiCoPO₄ (LCP) and LiFePO₄ (LFP).

Over the last decade, electrochemical properties of ILs were investigated by many groups [33, 34, 43–45]. Most of the imidazolium-based ILs suffer from a narrow ESW, which results in a poor cycling performance of the cell. This is due to low cathodic stability, which makes this type of RTILs inappropriate for applications in graphite-based lithium ion systems and to the formation of an ineffective SEI layer on the graphite surface, which results in the phenomenon of solvent intercalation and resulting into exfoliation, i.e., severe damage to graphite anodes by shedding of graphene layers or graphene layer packages [46]. However, imidazolium-based ILs have lower viscosities and higher conductivities as effect of the charge delocalization in the cations. Most importantly, the electrochemical performance of solvent blended electrolytes was investigated using different electrode materials. Guerfi et al. studied electrochemical properties of blended carbonate solvents with EMIm-TFSI on LFP cathode and mesocarbon microbeads (MCMB) anodes [29]. Although no solvent intercalation occurred at potentials lower than 0.25 V vs. Li/Li⁺, exfoliation of graphite was observed and the first cycle efficiency was very low when pure EMIm-TFSI-based electrolyte was used. The addition of carbonate solvents to EMIm-TFSI could improve the formation of an effective SEI layer due to the EC decomposition at ≈ 0.8 V vs. Li/Li⁺. The addition of an IL to the carbonate solvent clearly affected the first cycle Coulombic efficiency of the investigated electrolytes. The same was valid for first and second cycle capacity values, as summarized in Table 3.

The use of pure EMIm-TFSI and a LFP cathode resulted in lower first cycle Coulombic efficiencies, which could be due to high viscosity and low wettability of pure IL. Furthermore, the authors studied 40% EMIm-TFSI blends of 1 M LiPF₆ in

Table 3 The first cycle electrochemical characteristics of MCMB graphite anode in carbonate-based electrolyte blends and EMIm TFSI. Reprinted from Ref. [29]. Copyright (2010) with permission of Elsevier

IL (wt%)	Discharge cap. 1st/2nd	Charge cap. 1st/2nd	Ah. eff. 1st/2nd
100	141/94	96/79	68/84
50	311/255	250/248	80/97
30	377/314	315/312	85/99
20	388/324	337/331	87/102
10	362/306	329/302	91/99
0	362/306	316/312	92/102

EC/DEC (1:1 by wt.) and 2% VC as nonflammable electrolyte in a cell using LFP and LTO as cathode and anode. The formation cycle at C/24 rate had 89% Coulombic efficiency in the first cycle and a specific capacity of 115 mAh g⁻¹.

Kühnel et al. investigated the ESW of PC+ Py₁₄-TFSI solvent blended electrolytes with different concentrations of 0.3 M LiTFSI using Pt as working electrode [35]. No difference in electrochemical stability was observed in the blended electrolytes. The discharge capacities of all electrolytes with different concentrations were investigated in a half-cell setup using LFP as working electrode and C-rates between 0.1C and 5C. There is a good correlation between the percent amount of IL and the capacity; following the trend: the higher the IL amount, the lower the capacity (Fig. 9a).

Conductivity and wettability were found as two important factors influencing discharge capacity values of the nonaqueous electrolytes as claimed by the authors. The effect of electrode wetting on discharge capacities in galvanostatic cycling of blended electrolytes through the initial cycles is known in literature [47].

The performance of blended solvent electrolytes was investigated at high temperature (60 °C) by means of constant current cycling at 5 C. As an interesting

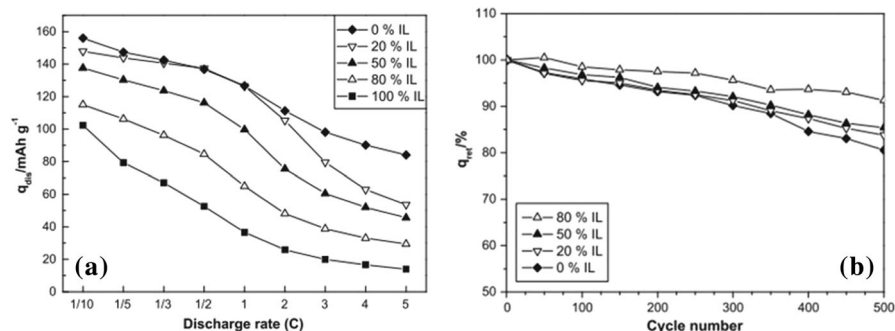
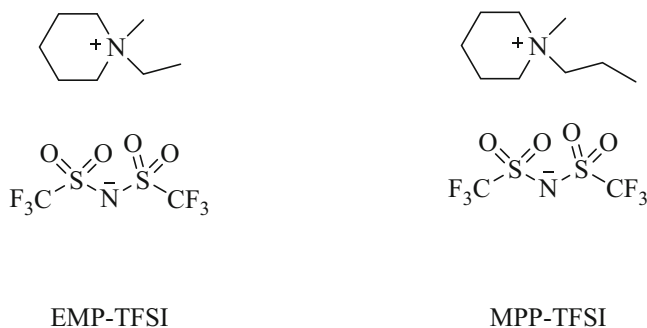


Fig. 9 **a** Specific discharge capacity (q_{dis}) of LFP electrodes with PC/Py₁₄-TFSI, 0.3 M LiTFSI blend electrolytes. The electrodes were cycled at RT with different discharge rates. Reprinted from Ref. [35]. Copyright (2011) with permission of Elsevier; **b** discharge capacity retention (q_{ret}) of LFP with PC/Py₁₄-TFSI, 0.3 M LiTFSI blend electrolytes. The electrodes were cycled at 60 °C with 5 C charge/discharge rate. Reprinted from Ref. [35]. Copyright (2011) with permission of Elsevier



Scheme 3 Molecular structures of 1-ethyl-1-methylpiperidinium bis(trifluoromethylsulfonyl)imide (EMP-TFSI) and 1-methyl-1-propyl-piperidinium bis(trifluoromethylsulfonyl)imide (MPP-TFSI)

result, the higher capacity retention for electrolytes with higher IL amounts during cycling was obtained (Fig. 9b). Kühnel et al. claimed that the lower capacity loss during cycling with higher amounts of Py₁₄-TFSI could be attributed to less dendrite formation or less solvent evaporation.

Kim et al. [48] published a paper on the high-voltage application of two different piperidinium-based ILs, namely 1-ethyl-1-methylpiperidinium bis(trifluoromethylsulfonyl)imide (EMP-TFSI) and 1-methyl-1-propyl-piperidinium bis(trifluoromethylsulfonyl)imide (MPP-TFSI) (molecular structures depicted in Scheme 3) in carbonate-based solvent using LMNO cathode.

Conductivity measurements revealed higher conductivity of the EMP-TFSI-based electrolyte with 1.5 M LiPF₆ in EC/DEC (1:1 v/v) compared to MPP-TFSI-based electrolytes. This has an obvious effect on galvanostatic cycling at different C-rates as the electrolyte with the lowest conductivity (MPP-TFSI blended electrolyte) had the largest over-potential, lower capacities as well as Coulombic efficiencies (Fig. 10).

Charge/discharge cycling at 0.5 C demonstrated very steady performance for both pure carbonate-based electrolyte and EMP-TFSI blended electrolyte, but MPP-TFSI blended electrolyte displayed very unstable cycling stability and Coulombic efficiencies over 50 cycles (Fig. 11).

2.3.2 Aluminum Dissolution in Presence of Nitriles

In electrolytes containing LiTFSI as conducting salt, anodic aluminum (Al) dissolution takes place [49–51]. Krämer et al. [52] showed that a blend of nitrile and carbonate slows down Al dissolution in LiTFSI-based electrolyte. The authors investigated dissolution behavior of Al at high potentials of 1 M LiTFSI-based [EC/EMC (1:1 by wt.), EC/DMC (1:1 by wt.), EC/DEC (1:1 by wt.), EC/DEC (3:7 by wt.)], pure ADN and EC/ADN (1:1 by wt.) electrolytes. They performed linear sweep voltammetry (LSV) measurements with a scan rate of 1 mV/s until a potential of 5.0 V vs. Li/Li⁺ was reached and thereafter kept constant for 24 h. The carbonate-based electrolytes showed a much higher electric charge (≈ 14 C after 24 h) whereas the EC/ADN blend led to an electric charge of ≈ 7.5 C. As

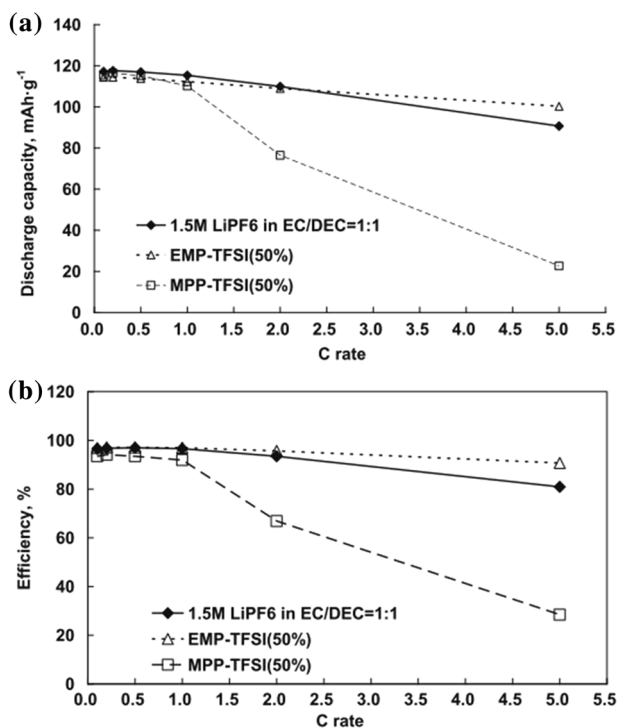


Fig. 10 The rate capability tests with 50% ILs and a pure carbonate electrolyte using LMNO working electrodes. **a** The discharge values versus the discharge rate and **b** the Coulombic efficiency versus the discharge rate. Prior to the measurements, five formation cycles at a rate of 0.1 C were performed for each measurement. Reprinted from Ref. [48]. Copyright (2013) with permission of Elsevier

comparison, 1 M LiTFSI in ADN as single solvent only reached an electric charge of ≈ 2.0 C. The dissolution of the current collector caused a weight loss, which was detected by weighting the Al electrode before and after the potentiostatic experiment. With pure ADN, no significant weight loss was observed, indicating negligible Al dissolution. The investigated EC/ADN blend led to a much higher weight losses, whereas carbonate-based electrolytes showed the highest weight loss and therefore the largest advance towards anodic Al dissolution. The fact that the anodic Al dissolution was slowed down, but remained apparent in EC/ADN blends with 1 M LiTFSI, led to the conclusion that LiTFSI is not a suitable conducting salt for EC/dinitrile blends. As LiPF₆ is soluble in ADN and GLN (<1 M) only in small quantities, even after adding carbonates, Isken et al. used LiBF₄ as conducting salt [31]. They revealed that 0.9 M LiBF₄ in EC/ADN (1:1 by wt.) electrolyte forms a stable protective film on the Al current collector, which prevents Al dissolution at 4.3 V vs. Li/Li⁺.

This issue was further investigated by Abu-Lebdeh et al. [15] in a 1 M LiTFSI in EC/ADN blend (1:1 v/v). They claimed that a higher open-circuit potential (OCP) indicates improved ability of the electrolyte to passivate Al surface. In order to further strengthen their idea, they employed the cyclic voltammetry (CV) technique

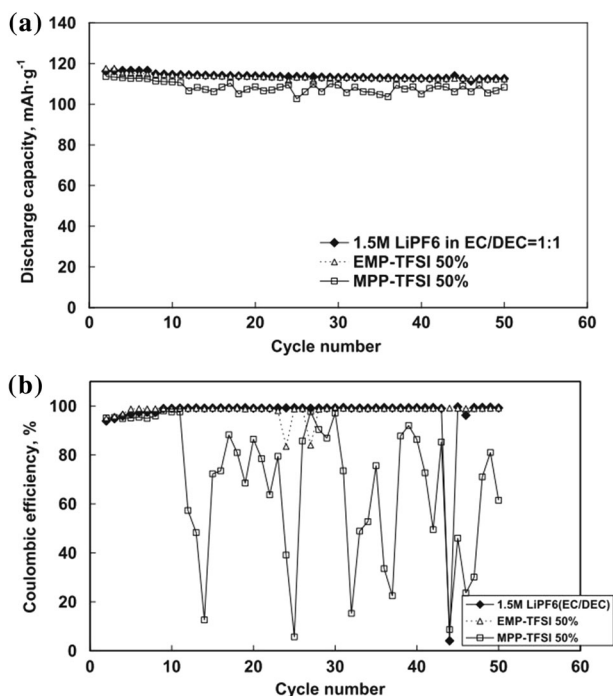


Fig. 11 Cycle life studies with IL blended electrolytes and a pure carbonate electrolyte using LMNO working electrodes. Both the charge and discharge rates for experiments were 0.5 C. Reprinted from Ref. [48]. Copyright (2013) with permission of Elsevier

in a three-electrode configuration with an Al wire as working electrode and a silver wire as a reference and counter electrode. The OCP of the cell containing the EC/ADN blend as solvent was reported as 2.51 V vs. Li/Li^+ which is higher than that of the cell with pure ADN (2.2 V vs. Li/Li^+). In light of these results, it was suggested that the solvent blend-based electrolyte formed a better passivation layer on Al. The passivation ability of the blended solvent electrolyte was better than that of 1 M LiTFSI in EC/DMC (1.63 V vs. Li/Li^+) [53]. Furthermore, by exchanging ADN with GLN, Abu-Lebdeh et al. [16] observed an OCP of 2.2 and 2.3 V vs. Li/Li^+ for pure GLN and EC/GLN blend (1:1 v/v), respectively, which is again higher than for the carbonate-based electrolyte.

Kühnel et al. has also investigated the anodic dissolution of Al current collector in electrolyte blends containing PC with different concentrations of 0.3 M LiTFSI in Py_{14} -TFSI. They used Al disks as working electrodes and lithium metal as counter and reference electrode to perform LSV measurements. It is well known that anodic dissolution of Al current collector is less pronounced in ionic liquid media probably due to low solubility of Al(TFSI)_3 species [54–56]. Here, the authors showed a correlation between the IL amount in solvent blended electrolyte formulations and anodic dissolution potentials. The values for blended electrolytes lay in between as listed in Table 4. The potential ranged from 3.8 V vs. Li/Li^+ in case of pure PC

Table 4 Maximum potentials ($E_{\max}$) applied to Al foil electrode in electrolytes before the Faradaic reactions start. Reprinted from Ref. [35]. Copyright (2011) with permission of Elsevier

IL (wt%)	$E_{\max}$ vs. (Li/Li ⁺) (V)
0	3.8
20	3.8
50	4.1
80	4.2
100	5.4

electrolyte with 0.3 M LiTFSI up to 5.4 V vs. Li/Li⁺ in case of pure Py₁₄-TFSI. For electrolytes with higher amount of IL, maximum potentials applied prior to Al anodic dissolution were higher than potential values in case of electrolytes with lower amount of IL.

2.4 Summary

Blending nitriles and ILs with carbonates enables their application as electrolyte co-solvents. In this way, better physicochemical and electrochemical properties required for the performance in a LIB cell can be achieved. Nitrile/carbonate solvent blends have lower melting points than the single solvents as well as higher flash points, thus leading to improved safety conditions. In addition, due to suppression of anodic Al dissolution and a broader electrochemical stability window, good electrochemical performance can be attained. Blending of RTILs with carbonate solvents influences the safety properties of the nonaqueous electrolytes. The addition of EC to the dinitrile and/or increase of the temperature lead to higher conductivity values. The capacity values are higher compared to pure dinitrile as solvent, but a solvent blend itself is not enough to insure adequate cycling performance. For this reason, a combination with a salt blend is required. Blends of RTILs and carbonate solvents contribute to more effective SEI layer formation on graphite electrodes and improve the conductivity and rate capability of the electrolyte. However, the addition of carbonate solvents to RTILs has a negative influence on safety properties of the electrolyte blend such as lower thermal stability and higher flammability. In summary, taking into account all investigated functional aspects, the combination of nitriles and RTILs with carbonates seems to be a good solution to overcome the drawbacks of the respective single solvents, as summarized in Table 5.

3 Blended Conducting Salts in Nonaqueous Electrolytes

All commercial nonaqueous LIB electrolytes require a conducting salt that, in the ideal case, fulfills the following requirements: high purity, low molecular weight, the ability to completely dissolve and dissociate in a nonaqueous solvent medium, intrinsic thermal stability, chemical stability vs. the solvent, stability over a wide electrochemical potential window, high mobility of the solvated ions (especially solvated Li⁺ ions), effective SEI and CEI formation on graphite and on the cathode

Table 5 Major roles of discussed solvent blend-based electrolytes

Solvent blend	Target improvement	Investigated electrolyte	References
EC/dinitrile	Higher flash point	1 M LiBF ₄ in EC/ADN (1:1 by wt.)	[31]
	Lower vapor pressure		
	Lower viscosity	1 M LiTFSI in EC/dinitrile (1:1 v/v)	[32]
	Higher conductivity	(NC-(CH ₂) _n -CN with <i>n</i> = 3–8)	
		1 M LiTFSI in EC/GLN (1:1 v/v)	[16]
		0.9 M LiBF ₄ in EC/ADN (1:1 by wt.)	[31]
	Suppressed Al dissolution	1 M LiTFSI in EC/ADN (1:1 by wt.)	[52]
	Effective SEI formation	0.9 M LiBF ₄ in EC/ADN (1:1 by wt.)	[31]
		1 M LiTFSI in EC/ADN (1:1 v/v)	[15]
		1 M LiTFSI in EC/GLN (1:1 v/v)	[16]
	Broader electrochemical stability window	0.9 M LiBF ₄ in EC/ADN (1:1 by wt.)	[31]
	Higher capacities	1 M LiTFSI in EC/ADN (1:1 v/v)	[15]
	Reduced capacity fading	1 M LiTFSI, 0.1 M LiBOB in EC/ADN (1:1 v/v)	
		0.9 M LiBF ₄ in EC/ADN (1:1 by wt.)	[31]
EC/DMC/dinitrile	Lower viscosity	1 M LiBF ₄ , 0.1 M LiBOB in EC/DMC/dinitrile (1:1:2 v/v/v)	[32]
	Higher conductivity		
	Higher capacities	1 M LiBF ₄ , 0.1 M LiBOB in EC/DMC/dinitrile (1:1:2 v/v/v)	[32]
		(NC-(CH ₂) _n -CN with <i>n</i> = 1–8)	
	Broader electrochemical stability window	1 M LiBF ₄ in EC/DMC/SEN (1:1:2 v/v/v).	[17]
EMIm-TFSI in EC/DEC (1:1 wt%)	Flame inhibition effect	EMIm-TFSI (40% or more) + 1 M LiPF ₆ in EC/DEC	[29]
	Optimum viscosity and conductivity values	EMIm-TFSI (40–60%) + 1 M LiPF ₆ in EC/DEC	
	Optimum charge/discharge capacity (MCMB/Li half-cell)	EMIm-TFSI (20–30%) + 1 M LiPF ₆ in EC/DEC	

Table 5 continued

Solvent blend	Target improvement	Investigated electrolyte	References
Pyr ₁₄ -TFSI in EC/DMC (1:1 wt%)	Flame inhibition effect	Pyr ₁₄ -TFSI (30% or more) + 1 M LiPF ₆ in EC/DMC	[30]
	Higher thermal stability	Pyr ₁₄ -TFSI + 1 M LiPF ₆ in EC/DMC	
Pyr ₁₄ -TFSI in PC	Higher anodic dissolution of Al current collector	PC+ 0.3 M LiTFSI in Pyr ₁₄ -TFSI (50% or more)	[35]
	Lower discharge capacities	PC+ 0.3 M LiTFSI in Pyr ₁₄ -TFSI (by addition of IL)	
	Higher capacity retention during galvanostatic cycling at 60 °C (LFP, 5C)	PC+ 0.3 M LiTFSI in Pyr ₁₄ -TFSI (by addition of IL)	
EMP-TFSI MPP-TFSI in EC/DEC (1:1 wt%)	Higher conductivity and better rate capability of EMP than MPP	50% IL (EMP or MPP) + 1.5 M LiPF ₆ in EC/DEC	[48]
	Improved galvanostatic cycling performance and higher Coulombic efficiency of EMP than MPP	50% IL (EMP or MPP) +1.5 M LiPF ₆ in EC/DEC	

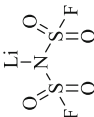
surface, passivation of the Al current collector to circumvent anodic Al dissolution, as well as low cost. Over the past three decades, various lithium salts (e.g. LiPF₆, LiBF₄, LiBOB, lithium difluoro(oxalato)borate—LiDFOB, LiTFSI, lithium bis(fluoro-sulfonyl)imide—LiFSI) were subjected to an extensive investigation in nonaqueous (aprotic) solvents or solvent blends in respect to their physicochemical, electrochemical and safety features [42]. Nevertheless, a lithium salt, which fulfills all the above demanded criteria at the same time, has not been identified so far. Moreover, compared to the wide variety of aprotic organic solvents available, the selection of an appropriate lithium salt is rather limited. Table 6 summarizes the most investigated lithium salts accompanied by their main physicochemical features.

The commercially dominant conducting salt in carbonate-based electrolytes is still LiPF₆. This domination has not been driven by any single outstanding property, but by the combination of a series of well-balanced properties [24]. However, its thermal and chemical instability has drawn the attention of the LIB society to search for alternatives [65]. In particular, the tendency of LiPF₆ to form toxic and chemically aggressive HF in the presence of water (as impurity in the LIB cell and the electrolyte itself) [66], which is accelerated by heat are a concern. LiPF₆ therefore cannot only deteriorate electrochemical performance, by e.g. increased cathode dissolution [67, 68], but can furthermore react with the organic solvents yielding even more toxic components such as fluorophosphates and organophosphates [5, 69–71].

Table 6 Main characteristics of the lithium salts discussed in the paper

Conducting salt	Molecular weight	Melting temperature (°C)	Decomposition temperature (°C) in solution	Al dissolution	Ionic conductivity (mS cm ⁻¹)	References
$\left[\begin{array}{c} \text{F} \\ \\ \text{F}-\text{P}-\text{F} \\ \quad \\ \text{F} \quad \text{F} \end{array} \right]^{-} \text{Li}^{+}$ LiPF₆	151.9	200	80 in EC/DMC	No	10.7 in EC/DMC	[24, 42, 57]
$\left[\begin{array}{c} \text{F} \\ \\ \text{F}-\text{B}-\text{F} \\ \quad \\ \text{F} \quad \text{F} \end{array} \right]^{-} \text{Li}^{+}$ LiBF₄	93.9	293	>100	No	4.9 in EC/DMC	[24, 25, 42]
$\left[\begin{array}{c} \text{O} \quad \text{O} \\ \diagdown \quad \diagup \\ \text{O} \quad \text{B} \quad \text{O} \\ \diagup \quad \diagdown \\ \text{O} \quad \text{O} \end{array} \right]^{-} \text{Li}^{+}$ LiBF₄	193.8	300	300–370	No	4.7 in EC/DMC	[25, 58–60]
$\left[\begin{array}{c} \text{O} \quad \text{O} \\ \diagdown \quad \diagup \\ \text{O} \quad \text{B} \quad \text{O} \\ \diagup \quad \diagdown \\ \text{O} \quad \text{O} \end{array} \right]^{-} \text{Li}^{+}$ LiBOB	143.8	265–271	240	No	4.5 in EC:DEC	[24, 61, 62]
$\begin{array}{c} \text{Li} \\ \\ \text{O} \quad \text{N} \quad \text{O} \\ \diagdown \quad \diagup \\ \text{F}_3\text{C} \quad \text{C} \quad \text{CF}_3 \\ \quad \\ \text{O} \quad \text{O} \end{array}$ LiDFOB	286.9	234	>100	Yes	9.0 in EC:DMC	[24, 57, 63]
LiTFSI						

Table 6 continued

Conducting salt	Molecular weight	Melting temperature (°C)	Decomposition temperature (°C) in solution	Al dissolution	Ionic conductivity (mS cm ⁻¹)	References
 LiFSI	187.1	135	>200	Yes	9.8 in EC:EMC	[24, 64]

3.1 Aim of Blending Lithium Salts

The electrochemical performance and safe operation of a battery cell mainly depend on the chemical and electrochemical stability of each individual cell component. Lithium salts, being responsible for (Li^+) ion conduction, may have strengths and weaknesses with respect to physicochemical and electrochemical features, which most likely restrict their application in future LIB systems. Substitution of LiPF_6 with new salts, addition of salt additives, or utilization of salt blends, is expected to improve electrolyte performance. With respect to salt blends, the known advantages of the individual conducting salts can be exploited by combining them. Salt blends exhibit improved thermal stability, ionic conductivity, electrochemical performance and safety characteristics as compared to their single constituents. Many interesting literature examples of blended salts that mainly shed light on the effect of salt blends on anode, cathode, suppression or inhibition of Al dissolution as well as the overall system performance were reported so far [72, 73]. The idea of blending various lithium salts in different amounts is generating considerable interest for future electrolyte formulation in LIB research. However, there is still much controversy surrounding the influence of salt blends on electrolyte properties. In addition, a common approach in improving electrolyte formulations is related to the addition of the sacrificial anions, so-called salt additives, which degradation results in a considerable improvement of battery performance [74]. By combining the advantages of two different salts, cell chemistry and cell performance can be optimized and tailored for a specific purpose. Lithium salts were used as blends in various concentrations and combinations to suppress the adverse effects of the single salt in electrolyte formulation. The purpose of blending salts can be listed as (1) achieving higher ionic conductivity by combining several lithium salts in one pot, (2) adjusting physicochemical features of the resulting electrolyte blends, e.g. boiling, freezing points, solubility, (3) forming an effective SEI on the surface of the negative electrode, thus improving its cycling efficiency in lithium ion or lithium metal batteries, (4) preventing metal dissolution from the cathode, (5) increasing anodic potential stability window, (6) forming a passivation layer on the surface of the Al current collector and preventing anodic Al dissolution and (7) improving cell safety.

3.2 Physicochemical Features of Blended Salt-Based Electrolytes

3.2.1 Transport Properties: Viscosity and Conductivity

The viscosities of $\text{LiPF}_6 + \text{LiBOB} + \text{LiDFOB}$ in EC/DMC (1:1 by wt.) and EC/dimethyl sulfone (DMSN) (4:1 by wt.) were systematically investigated by Hofmann et al. [75]. Their findings indicated an increase in viscosity and a decrease in ionic conductivity for the following binary and ternary blended salts: $\text{LiPF}_6 + \text{LiBOB}$, $\text{LiPF}_6 + \text{LiDFOB}$ and $\text{LiPF}_6 + \text{LiBOB} + \text{LiDFOB}$, as compared to the single-salt system (LiPF_6). As summarized in Table 7, no significant difference could be observed.

Table 7 Physicochemical features of binary and ternary blended salts over single salt by Hofmann et al. Reprinted from Ref. [75]. Copyright (2015) with permission of Elsevier

Electrolyte	LiPF ₆ EC/DMC	LiPF ₆ EC/DMSN	LiPF ₆ + LiBOB EC/DMSN	LiPF ₆ + LiDFOB EC/DMSN	LiPF ₆ + LiBOB + LiDFOB EC/DMSN
Melting point (°C)	−19.4	16.0	18.1	18.6	18.8
Freezing point (°C)	−56.1	−26.1	−12.3	−19.1	−17.3
Density (25 °C) (g cm ^{−3})	1.27	1.40	1.39	1.36	1.37
Viscosity (20 °C) (mPas)	4.4	11.4	11.0	11.4	11.4
Conductivity (mS cm ^{−1})	10.67	5.95	5.38	5.95	5.76

The physicochemical features of a series of electrolytes containing LiBF_4 + LiBOB salt blends were reported by Lai et al. with an attempt to improve the operation temperature range of LIBs. In their study, the investigated electrolyte formulations contained salt blends of LiBF_4 + LiBOB (8:2) in PC/EC/EMC, PC/EC/EMC/MB, PC/EC/methyl butyrate (MB) solvent systems [76]. At moderate and high temperatures, ionic conductivity values of the blended salt electrolytes were lower than that of single component salt (LiPF_6). However, at lower temperatures (-40 and -20 °C), no considerable difference could be observed due to the high mobility of BF_4^- . Electrolyte blends of LiBF_4 + LiBOB (8:2) PC/EC/EMC/MB (1:1:1:2 v/v/v/v) showed ionic conductivity of 4.34 and 2.16 mS cm^{-1} at -20 and -40 °C, respectively.

The discrepancy in ionic conductivities of LiBF_4 + LiBOB salt blends as compared to the single constituents and LiDFOB salts was discussed by Schedbauer et al. [62]. The specific conductivities of the LiBF_4 containing electrolyte blends were found to be lower as compared to non- BF_4^- containing electrolytes in the temperature ranging from -40 to 60 °C. Moreover, 0.5 M LiBOB in EC/DEC (3:7 by wt.) and 1 M LiDFOB in EC/DEC (3:7 by wt.) electrolytes showed almost identical behavior in the investigated temperature range. Furthermore, the trend in conductivities of LiBOB + LiBF_4 salt blends depicted a descending behavior with the increasing LiBF_4 concentration.

In the study published by Zhang et al., the effect of the increasing LiFSI concentration on the electrochemical performance of LiPF_6 -based EC/EMC (3:7 v/v) electrolyte was discussed [77]. Ionic conductivity of the resulting blend showed a parallel trend with the concentration of LiFSI up to 0.2 mol L^{-1} ($\sigma_{0.1 \text{ mol L}^{-1}}$: 9.33 mS cm^{-1} , $\sigma_{0.2 \text{ mol L}^{-1}}$: 9.60 mS cm^{-1}). However, at concentrations higher than 0.2 mol L^{-1} a fast decrease in ionic conductivity of the electrolyte was observed, which could be attributed to the detrimental effect of the high salt concentration, giving rise to ionic interactions.

A similar trend was observed for LiBF_4 + LiDFOB salt blends in EC/EMC/DMC (1:1:1 by wt.) [78]. Concentration changes of LiDFOB and LiBF_4 were found to influence the ionic conductivity in a different manner. An increase of the LiDFOB concentration (up to 0.8 M) led to higher ionic conductivities in the temperature range from 30 to 70 °C compared to that of the pure LiBF_4 -based electrolyte [79, 80]. At contrast, in the temperature range from 20 to -20 °C, the ionic conductivity of the aforementioned system gradually deteriorated. However, an increase of the LiBF_4 concentration (up to 0.8 M) in this low temperature range had a positive influence on ionic conductivity, due to the lower van der Waals volume of the BF_4^- ion and its correlation to electrolyte viscosity [81]. An equimolar blend of LiBF_4 and LiDFOB in EC/PC/DMC (1:1:3) at a concentration of 10 wt%, provided a good overall ionic conductivity (in the order of $10^{-3} \text{ S cm}^{-1}$) over the temperature range from 20 to 80 °C as stated by Zygadło-Monikowska et al. [82]. Besides, a LiPF_6 + LiDFOB blend in EC/PC/DMC (1:1:3 v/v/v) could not compete with the pure LiPF_6 -based electrolyte in the temperature range from 0 to 40 °C, as showed by Zhang et al. [83]. This is due to the larger van der Waals volume of the ODFB^- ion compared to PF_6^- as well as the lower ionic dissociation

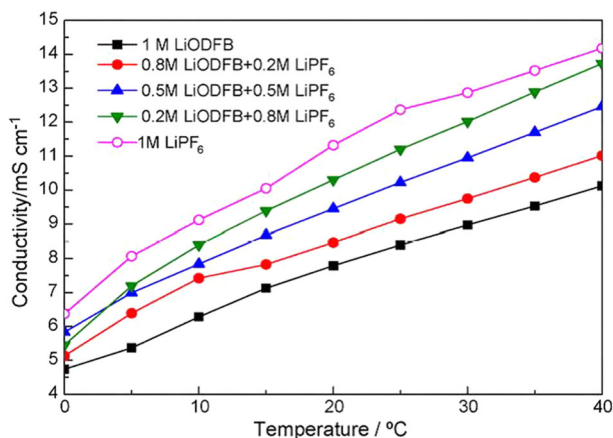


Fig. 12 Ionic conductivity (κ) of electrolytes with different LiDFOB + LiPF₆ salt blend ratios in EC/PC/DMC (1:1:3) solution (0–40 °C). Reprinted from Ref. [83]. Copyright (2010) with permission of Elsevier

ability of LiDFOB compared to LiPF₆, leading to higher viscosity of the electrolyte, thus resulting in conductivity decrease. With this in mind, the conductivity of the LiDFOB-based electrolytes could be enhanced by addition of LiPF₆, as depicted in Fig. 12 [83].

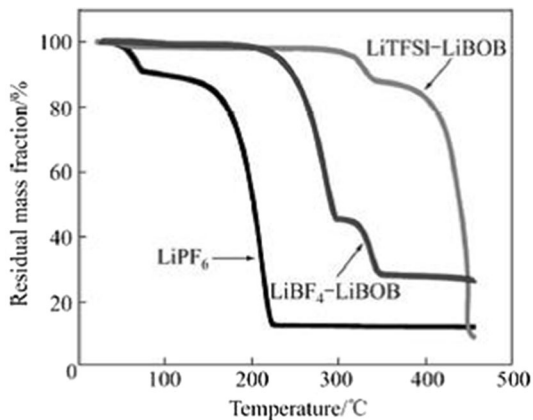
In another study by Park et al., the ionic conductivity of (EC/DEC 3:7 v/v) electrolyte containing blend of LiFSI and LiDFOB salts was determined over the temperature range from 10 to 60 °C and compared to the pure LiFSI and LiPF₆-based electrolytes [84]. In consistence with literature data, the conductivity of the LiFSI-based electrolyte was higher than that of LiPF₆-based electrolyte [64, 85]. In line with this, the ionic conductivity of the 0.8 M LiFSI + 0.2 M LiDFOB in EC/DEC (3:7 v/v) was found to be higher than the pure LiPF₆-based electrolyte over the aforementioned temperature range indicating that addition of LiDFOB had no significant impact on the superior ionic conductivity of the pure LiFSI-based electrolyte [84].

Using the synergistic effect of LiDFOB and LiTFSI salts in EC/EMC (4:6 by wt.), Li et al. pointed out the minor difference in ionic conductivity of equimolar blend (7 mS cm⁻¹) compared to the pure LiTFSI-based electrolyte (7.8 mS cm⁻¹) at 20 °C [86].

3.2.2 Thermal Stability of Blended Salt-Based Electrolytes

It is of great importance to introduce thermally stable electrolyte blends for achieving risk-free operation of LIBs. Several approaches aimed at suppressing the low thermal stability of the LiPF₆-based electrolyte by using thermally more stable salt blends including LiBOB, LiBF₄, LiDFOB and LiTFSI. Furthermore, in few studies LiBOB + LiPF₆ salt blends were used for elucidating the effect of LiBOB on suppressing the thermal instability of LiPF₆ salt [72, 73]. The influence of LiBOB on the thermal stability of MCMB/spinel LIBs was attained from the

Fig. 13 TGA profiles of LiBOB + LiTFSI and LiBOB + LiBF₄ salt blends compared to pure LiPF₆. Reprinted from Ref. [87] Copyright (2015) with permission of Elsevier



corresponding cycling curve at 55 °C, which depicted more steady behavior as compared to the pure LiPF₆ salt-based electrolyte.

Zhang et al. [87] reported several thermally stable salt blends, namely LiBF₄ + LiBOB and LiTFSI + LiBOB as compared to the SOTA salt, LiPF₆. The thermal stability of the aforementioned blended salt electrolytes is illustrated in Fig. 13.

Another example related to the effect of blending on the thermal stability for LiBF₄ containing electrolyte was disclosed by Hong et al. [88]. As already mentioned, LiBF₄ suffers from low ionic conductivity but it provides good cycling performance at higher (50–80 °C) and lower temperatures (–20 °C) [89]. A study published by Zhang et al. pointed out that blending LiBOB salt with LiBF₄ and LiTFSI enables effective SEI formation and Al current collector passivation, which can successfully compensate the drawbacks of LiBF₄ and LiTFSI. According to their results, the thermal stabilities of the investigated salt systems followed the ascending order of LiPF₆ < LiBF₄ + LiBOB < LiTFSI + LiBOB [89, 90].

By blending LiPF₆ with LiDFOB in EC/PC/DMC (1:1:3 v/v/v), Zhang et al. showed remarkable improvement in performance of the artificial graphite (AG)/LFP cells at 65 °C compared to the pure LiPF₆-based electrolyte formulation. This superior behavior, expressed through higher capacity and capacity retention values, was attributed to an effective SEI film-formation in the presence of the high LiDFOB concentration (LiDFOB/LiPF₆ in ratio 4:1) as well as the thermal stability enhancement of the electrolyte. The presence of LiDFOB inhibited the autocatalytic decomposition of the electrolyte and synergistic effect of both salts enhanced the thermal stability of the electrolyte. This was achieved by a disproportionation reaction of LiDFOB, in which lithium tetrafluorooxalatophosphate (LiPF₄C₂O₄), which sequestered PF₅ formed during thermal decomposition of LiPF₆, was generated [91]. The addition of LiPF₆ in smaller amount contributed to an improved electrolyte conductivity and increased capacity [83]. The beneficial influence of LiDFOB + LiBF₄ blends in EC/DMC/EMC (1:1:3 by wt.) was confirmed in graphite/LiNi_{0.5}Mn_{1.5}O₄ (LNMO) full cells at low (–20 °C) and elevated (60 °C) temperatures [78]. Moreover, the improved low temperature performance at high

C-rate charging and discharging was attributed to a diminished charge transfer impedance in presence of LiBF_4 ($\text{LiDFOB}/\text{LiBF}_4 = 4:1$, in molar).

3.3 Electrochemical Features of Blended Salt-Based Electrolytes

3.3.1 Cathodic Stability of Blended Salt-Based Electrolytes

The cathodic stability of the blended salt-based electrolytes was mainly characterized by means of CV and/or galvanostatic charging on the surface of a graphite electrode. The performance assessment of such electrolyte blends was derived based on: (1) a SEI forming features, (2) irreversible capacity, (3) Coulombic efficiency and (4) capacity retention.

As initially emphasized, the blending concept significantly influences cell chemistry and electrochemical features of the electrolyte system. A well-known example is given for a PC containing electrolyte formulations in combination with graphitic anodes. The superior features of PC-based electrolytes such as enhanced rate capability, lower and higher temperature stability make this solvent an important component for electrolyte formulations [92]. Nevertheless, its incompatibility to the graphitic electrode, due to solvent co-intercalation and intense gas evolution surface is a well-known issue [93, 94]. In this respect, a blended salt system that contains sufficient amounts of LiBOB (1–5 mol%) in LiPF_6 or LiBF_4 -based electrolytes, may serve as a promising solution to cope with PC reduction and graphite exfoliation. Xu et al. investigated two electrolyte formulations that contained 50 and 30% PC, which in the normal case prevent intercalation of Li^+ into graphite. In the additive free case, the graphite potential leveled off at 0.8 V vs. Li/Li^+ which is in the characteristic reduction potential range of PC (Fig. 14). However, in the presence of LiBOB, the reversible lithiation potential of graphite (0.2 V vs. Li/Li^+) could be reached [58]. Another example for a PC containing electrolyte recipe was related to the $\text{LiBOB} + \text{LiBF}_4$ salt blend which was considered as solution to enable the application of a LiBF_4/PC -based electrolyte in graphite-based LIBs [95], by introducing LiBOB as a SEI forming additive. A cathodic signal at 1.75 V vs. Li/Li^+ indicated the reduction of BOB^- forming an

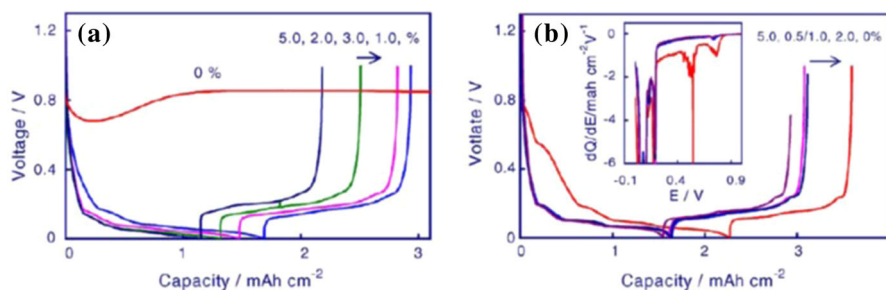


Fig. 14 Voltage-capacity curves of the first cycle for Li/graphite cells with different LiBF_4/PC electrolytes, in which the numbers present molar percentage of LiBOB in the mixed salt. *Inset in b* is a part of the plots of differential capacities vs. cell voltage. **a** 1.0 m LiBF_4 PC/EC (1:1) and **b** 1.0 m LiBF_4 PC/EC/EMC (1:1:3). Reprinted from Ref. [95]. Copyright (2006) with permission of Elsevier

effective SEI on the negative electrode [96, 97]. As reported by Panitz et al. reduction of LiBOB gave rise to capacity loss and higher charge-transfer impedance as compared to the SOTA salt [98]. Moreover, in the same study it was reported that the state of the graphite anode surface, e.g. surface chemistry [99–102], specific surface area [103–106], have a decisive role in the intensity of this reduction process. The opportunity to tailor SEI chemistry of a carbonaceous electrode by a boron rich surface layer made LiBOB a good electrolyte candidate for blending purposes.

A similar study reporting on the superiority of LiBOB with regard to the SEI film-formation, was performed by Zhang et al. [95]. When LiBOB was used as co-salt, it sufficiently stabilized the surface of the graphitic anode and in this way enabled the application of LiBF₄/PC electrolytes in LIBs. Additionally, it was indicated that LiBOB, when used even in small amount (1–5 mol%), formed an effective SEI film on the graphite electrode [107] in LiBF₄-based PC/EC/EMC and LiBF₄-based PC/EC electrolytes. The effectiveness of LiBOB in circumventing the solvent reduction at potential regimes of 0.8–0.4 V vs. Li/Li⁺ was investigated by means of the CV technique. In the absence of LiBOB, solvent reduction peaks within the indicated potential range were clearly observable, whereas in the presence of 1 mol% LiBOB the peaks were suppressed and Li⁺ intercalation into graphite could take place at ≈ 0.3 V vs. Li/Li⁺. Even at low concentrations of LiBOB (0.5 mol%), the SEI was sufficient to suppress further reduction of the investigated electrolyte. The superior feature of the BOB[−] originated SEI was reported to advance the thermal reactivity of the lithiated graphite towards the nonaqueous electrolyte and in this way, an improvement in the safety characteristics of the negative electrode could be observed. In a further study performed by Jiang et al., the thermal stability of the lithiated graphite electrodes was investigated by means of accelerated rate calorimetry (ARC). According to the obtained results, the exothermic reaction between lithiated MCMB and the electrolyte began at 80 °C in a LiPF₆ salt-based electrolyte system, whereas in the presence of LiBOB it took place at 180 °C [59, 108, 109].

An interesting study involving a LiPF₆ + LiFSI + LiBOB ternary salt blend for achieving improved electrochemical performance of MCMB/LFP full cells was conducted by Zhang et al. [77]. They indicated outstanding capacity retention (99.6% after 100 cycles) of an MCMB electrode in a 1 mol L^{−1} LiFSI-based EC/EMC (3:7 v/v) electrolyte which served as major reason to include LiFSI in the electrolyte; however, Al current collector dissolution in presence of the FSI[−] anion limits application of these electrolyte formulations. In fact, the authors presented that LFP-based cells could only be cycled for seven cycles in LiFSI + LiPF₆ salt-blended electrolyte (1 mol L^{−1} LiPF₆ and 0.2 mol L^{−1} LiFSI) due to the severe Al dissolution. For this reason, the addition of LiBOB salt into this binary blend to effectively passivate Al surface with an optimum concentration of LiPF₆ + LiFSI + LiBOB (0.8:0.2:0.2 mol L^{−1}) was proposed. The capacity retention values after 120 cycles in graphite-based cells containing LiPF₆, LiPF₆ + LiFSI, LiPF₆ + LiFSI + LiBOB as single, binary and ternary salt blends were reported to be 88.2, 93.5 and 97.0%, which was another proof for the effectiveness of the ternary blends on the cycling performance of graphite. Nevertheless, the rate

capability of the MCMB electrode in the $\text{LiPF}_6 + \text{LiFSI}$ salt blend-based electrolyte was slightly lower than that of the ternary blend ($\text{LiPF}_6 + \text{LiFSI} + \text{LiBOB}$), which could be explained by the higher SEI resistance caused by BOB.

Schedbauer et al. conducted a study on the composition of the SEI on copper surface in the presence of a $\text{LiBF}_4 + \text{LiBOB}$ salt blend-based electrolyte [62]. Results obtained by means of X-ray photoelectron spectroscopy (XPS) measurements showed that with increasing the LiBF_4 amount in the $\text{LiBF}_4 + \text{LiBOB}$ blend, the composition of the SEI displays similarity to that containing the LiDFOB salt. The interaction between LiBF_4 and LiBOB resulted in a trace amount of LiDFOB formation at room temperature as was confirmed by NMR measurements.

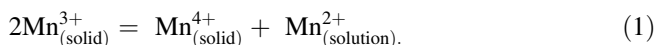
Cycling stability of lithium-metal batteries was limited by the lithium dendrite growth on lithium metal during repeated cycling which causes severe safety risks for commercial applications [110–113]. It is known that the SEI films on Li metal and lithiated graphite differ considerably [114, 115]. An electrolyte based on a $\text{LiTFSI} + \text{LiBOB}$ salt blend was investigated by Xiang et al. with an aim to improve the charging capability of $\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ (NCA)/Li half-cells [116]. In the investigated electrolyte system, LiBOB played a crucial role in forming an effective SEI film on Li metal and the mandatory Al current collector passivation, whereas LiTFSI functioned in modifying the features of the SEI film on the lithium metal by increasing ionic conductivity.

As stated by Zhang et al., blending of LiPF_6 and LiDFOB salts in a ternary carbonate-based electrolyte, contributed to improved performance of both artificial graphite (AG)/Li half- and AG/LFP 063048-type full cells [83]. Compared to a pure LiPF_6 -based electrolyte, in the presence of LiDFOB , a more effective SEI was formed in the initial cycles, resulting in a superior Coulombic efficiency of 98.2% in the third cycle and a reversible capacity of 339 mAh g^{-1} over 100 cycles in the AG/Li half-cell. The obtained CV results indicated a strong influence of LiDFOB on the Li^+ de-intercalation step. The optimized $\text{LiPF}_6/\text{LiDFOB}$ ratio was set to 1:4 (0.2 M:0.8 M). In the AG/LFP full cell setup, the obtained discharge capacity values were significantly higher compared to the pure LiPF_6 -based electrolyte. Furthermore, the presence of LiPF_6 in a blended salt system was found to compensate the poor ionic conductivity of the LiDFOB -based electrolyte. The authors even stated that it might additionally have a positive influence on the capacity values. After 100 cycles, the cell containing 0.2 M $\text{LiPF}_6 + 0.8 \text{ M LiDFOB}$ in EC/PC/DMC (1:1:3 v/v/v) electrolyte showed a capacity higher than the cell with the pure salt electrolyte. In addition, at elevated temperature (65°C), both discharge capacity and capacity retention of the optimized salt-blended electrolyte (351.3 mAh, 88.7%) were superior compared to the single salt containing electrolyte (121 mAh, 52%).

3.3.2 Anodic Stability of the Blended Salt-Based Electrolytes

In a study performed by Shieh et al., the effect of $\text{LiBOB} + \text{LiPF}_6$ salt blend on thermal stability, cycle life stability and Mn dissolution of LMO electrode was discussed [72]. Mn^{2+} dissolution from the LMO cathode material in the SOTA

electrolyte system (1 M LiPF₆ EC/DMC) at elevated temperatures and its consequences on cell chemistry are a well-known issue [117, 118]. At elevated temperature, decomposition of LiPF₆ triggers the Mn²⁺ dissolution from LMO via the Hunter's mechanism [119]:



Cycling investigations showed no significant difference in capacity fading of the single-salt containing systems (LiBOB or LiPF₆) with 79% capacity retention, whereas the capacity retention of the blended salt electrolytes (82% for 0.5 M LiBOB + 0.5 M LiPF₆) was reported to be superior to the single salt-based electrolyte after 100 cycles.

An electrolyte containing LiBOB + LiPF₆ salt blend, enabled a Mn²⁺ dissolution decrease from the LMO cathode thus stabilizing the LiPF₆ salt decomposition during cycling at elevated temperature as well as enhancing the electrochemical performance of the cell. In conclusion, the blended salt concept potentially allowed for application of LMO at elevated temperatures (>55 °C). A similar study, performed by Chen et al. [73], confirmed the findings of Shieh et al. [72]. Their results showed that the power capability of lithium ion cells decreases with increasing LiBOB concentration; however capacity retention of the cells at 55 °C increased with LiBOB concentration which is in line with the results obtained by Shieh et al. [52]. Furthermore, they specified the optimum concentration of LiBOB as 0.1 M, using MCMB/NMC full cells at 55 °C. MCMB/spinel full cell cycling at 55 °C with 0.1 C current rate showed a clear evidence of poor cycling behavior in the SOTA electrolyte formulation (1.2 M LiPF₆ in EC/PC/DMC) that led to more than 50% capacity loss in 300 cycles, most probably due to Mn²⁺ migration from the positive to the negative electrode and deterioration of the SEI. However, the pure LiBOB and blended (LiPF₆ + LiBOB) salt systems significantly improved the cycling stability of the cell, which pointed out an effective CEI formation and suppressed Mn²⁺ dissolution in presence of LiBOB.

A ternary salt blend containing LiPF₆ as main salt, with LiFSI and LiBOB as co-salts was investigated by Zhang et al. for improved electrochemical performance of a graphite/LFP full cell [77]. The aim of their study was to tailor the electrolyte by using LiFSI (high conductivity and good cycling stability) + LiBOB (beneficial Al current collector passivation behavior) blend. Graphite/LFP full cells cycled in this ternary salt blend-based electrolyte exhibited improved cycling characteristics; the Coulombic efficiency was close to 100% over 200 cycles.

LiFSI, as conducting salt, generated widespread interest in LIB science owing to its promising features, e.g. absence of hydrogen fluoride (HF) release, enabling good electrolyte conductivity, high-rate capability and low electrolyte viscosity [85, 120]. However, a key challenge is addressed to the Al current collector dissolution, thus limiting the application of this salt [121].

An electrolyte based on LiTFSI + LiBOB blend was suggested by Xiang et al. to improve the charging capabilities of NCA/Li half-cells [116]. In such electrolyte system, LiBOB takes part in formation of an effective SEI film and passivation of

the Al current collector, whereas LiTFSI functioned in modifying the SEI on lithium metal by increasing its ionic conductivity.

In view of the realization of high voltage LIBs, it is important to introduce electrochemically stable electrolyte systems that operate beyond 5 V vs. Li/Li⁺. Li et al. investigated a blended salt-based electrolyte formulation with an aim to improve the cycling stability of the LNMO cathode at high potentials [122]. In their study, LiBOB had a role of a main conducting salt, whereas lithium difluoro(-sulfato)borate (LiBSO₄F₂) was used as a co-salt in 0.1 M to create an effective CEI at the LNMO cathode surface. As suggested by the authors, LiBSO₄F₂ formed a smooth and uniform, thus low resistive CEI. Their findings indicated that the LiBSO₄F₂ + LiBOB-based electrolyte exhibited a lower charge-transfer impedance with LNMO due to the formation of a CEI, which was found to be rich in sulfurous compounds. The oxidation stability increased from 5.3 to 5.6 V vs. Li/Li⁺. Moreover, the influence of the co-salt was further proven by electrochemical impedance spectroscopy results showing that the LiBSO₄F₂ salt reduced the charge-transfer impedance. The capacity retention of LNMO electrodes in this blended salt system was reported to be 98.9% after 100 cycles, whereas only 92.8% retention could be achieved in the single salt electrolyte. Finally, the C-rate capability was improved with the salt blend-based electrolyte.

In a recent study, Zhou et al. indicated the LiDFOB + LiBF₄ salt blend as a promising candidate for enabling the high voltage LNMO cathode active material [78]. The authors reported enhanced electrochemical kinetics at room temperature of the LNMO electrodes in the electrolyte containing high amount of LiDFOB compared to the pure LiBF₄-based electrolyte as well as the strong influence of this salt on reaction reversibility and capacity retention of the investigated graphite/LNMO full cells. As a result of the effective CEI formation in presence of LiDFOB (LiDFOB/LiBF₄ ratio = 4:1), the specific capacity and capacity retention after 100 cycles amounted to 120.7 mAh g⁻¹ and 95.2%, respectively. SEM results of the pristine and cycled LNMO electrodes confirmed the formation of thin CEI layer in the presence of LiDFOB + LiBF₄ salt-blended electrolyte, whereas in case of the pure LiBF₄-containing electrolyte, continuous electrolyte decomposition led to a formation of thick and dense CEI layer which had a detrimental effect on the capacity retention at elevated temperature. In addition, the fluoroborate salts LiBF₄ and LiDFOB in an equimolar blend of 0.1 M in DMC showed an oxidation stability up to 4.9 V vs. Li/Li⁺ [82]. By combining the advantages of both conducting salts, the LiTFSI + LiDFOB blend in EC/EMC (4:6 by wt.)-based electrolyte formulation was found to significantly improve the electrochemical performance of the LFP electrode. The optimum amount was determined to be 1 M LiTFSI_{0.6} + LiDFOB_{0.4}. Although the increase of LiDFOB concentration improved the cycling performance, concentrations higher than 0.4 M would lead to the higher cell impedance. Figure 15 illustrates the voltage profiles of LFP/Li half-cells at different C-rates of the salt blend containing electrolyte compared to the pure LiPF₆-based electrolyte.

The reasons behind the improved Coulombic efficiencies were related to superior water and thermal stabilities of both the LiBF₄ and LiDFOB salt resulting in suppressed iron dissolution and weak parasitic reactions.

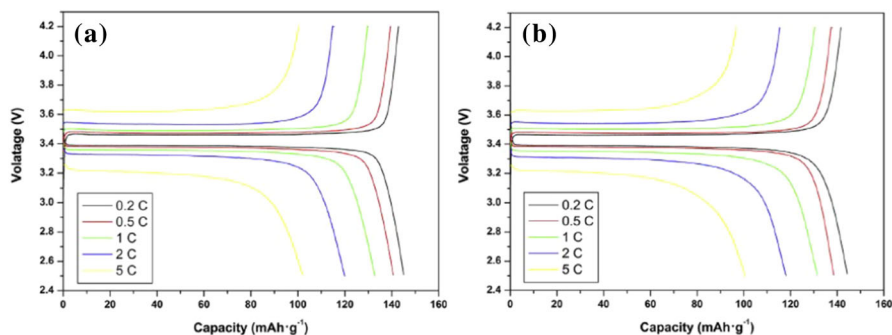


Fig. 15 Rate capability of LFP/Li half-cells in electrolytes containing **a** 1 M LiTFSI_{0.6} + LiODFB_{0.4} and **b** 1 M LiPF₆. Reprinted from Ref. [86]. Copyright (2015) with permission from Elsevier

3.3.3 C-Rate Capability and Capacity Retention of the Blended Salt-Based Electrolytes

As reported by Chen et al. [73], no considerable change in cycling stability between the investigated single and binary salts containing electrolytes was observed, whereas differences in capacity retention at elevated temperatures (55 °C) were clearly visible, as reflected from Fig. 16. The LiBOB containing electrolyte presented a better cycling performance in graphite/NMC full cells (10% capacity loss over 60 cycles in LiBOB free electrolyte). Similarly, a poor capacity retention was reported for the graphite/spinel full cells (50% loss in 300 cycles) at 55 °C. The ternary salt blend (LiPF₆ + LiBOB + LiFSI) exhibited a capacity retention close to 100% with a specific discharge capacity of 153.9 mAh g⁻¹ for graphite/LFP full cells [77].

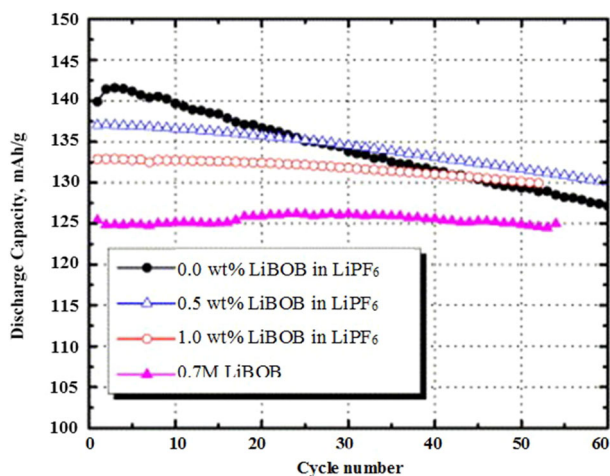


Fig. 16 Capacity retention of graphite/NMC (MCMB-1028/L333 cells at 55 °C). The cells were cycled with a constant current of C/2 (~1.2 mA). Redrawn from Ref. [73]. Copyright (2006) with permission of Elsevier

Cycling stabilities of graphite/LiNiO₂ cells containing LiBOB + LiBF₄ salt blend in PC/EC/EMC-based electrolyte were reported by Zhang et al. [95]. The results indicated that the cell maintained 90% of the initial capacity after 180 cycles at room temperature in 2 mol% LiBOB containing electrolyte formulation. Another study on LiBOB + LiPF₆ blends stated that the capacity loss for the blended salt-based electrolyte (0.75 M LiPF₆ + 0.25 M LiBOB) was 16%, whereas 23% was observed in the LiPF₆ case over 250 cycles at 55 °C [72].

The capacity retentions of several blended electrolyte systems containing LiBF₄ + LiBOB (8:2) in various solvent blends with LiPF₆ as reference were reported by Lai et al. [76]. Their findings indicated that at elevated temperature (65 °C) the LiPF₆-based electrolyte presented 77% capacity retention in LFP/Li half-cells, whereas LiBF₄ and LiBOB blends maintained the capacities at 79–85% of the initial value. In this respect, 1 M LiBF₄ + LiBOB (8:2) salt blend in EC/PC/EMC (1:1:3) was identified as the most promising electrolyte formulation. A similar study performed by Zhang et al. disclosed a capacity retention of the LiBF₄ + LiBOB salt blend for the LFP/Li half-cell as 87.5% as compared to LiTFSI + LiBOB with 85.1% and LiPF₆ with 79.3% over 100 cycles at 65 °C, [87]. With these findings in line, blended salt electrolytes demonstrated better capacity retention at high temperature as compared to the single-salt system.

3.3.4 Suppression of Aluminum Dissolution by Blended Salt-Based Electrolytes

To ensure safe and long-lasting performance, a compact protective passivation layer must be formed on the surface of the Al current collector. The following factors mainly determine the passivation process: the upper cut-off potential, the electrolyte solvents, additives, and the used conducting salt. The potential at which the oxidation dissolution current rapidly increases in reference to the background current is called 'pitting potential (E_p)'. Only at potentials higher than E_p , the oxidation currents vary with the conducting salt. The presence of LiBF₄ and LiPF₆ suppresses this current at higher potential values, thus enabling effective passivation of the Al surface and preventing further oxidation. On the contrary, lithium imide salts (LiIm) are known for promoting severe Al current collector dissolution [123, 124]. Among them, LiFSI was reported as promising salt to overcome the well-known limitations of LiPF₆ [120, 121]. With this in line, Park et al. systematically investigated the inhibition of dissolution in LiFSI in EC/DEC (3:7 v/v) electrolytes by addition of selected Li borate salts and revealed the following order of their inhibition ability: LiDFOB > LiBF₄ $\approx$ LiPF₆ > LiBOB [84]. Furthermore, a 0.8 M LiFSI + 0.2 M LiDFOB salt blend in EC/DEC (3:7 v/v) showed remarkable inhibition of Al dissolution, even comparable to a 1 M LiPF₆ electrolyte. As depicted in Fig. 17, the superior inhibition ability of LiDFOB, compared to other investigated borate salts, was assigned to the increased thickness (~ 24.4 nm) of the passivation surface layer compared to the layer in other blends (<1 nm) as well as to the nature of its constituents, Al–F, Al₂O₃ and B–O/B–F species [84]. B–O compounds are known for their ability to protect Al surface from dissolution caused by LiIm salt-based electrolytes [125]. LiDFOB was suggested as the most effective complementary salt to LiFSI for improving Al stability.

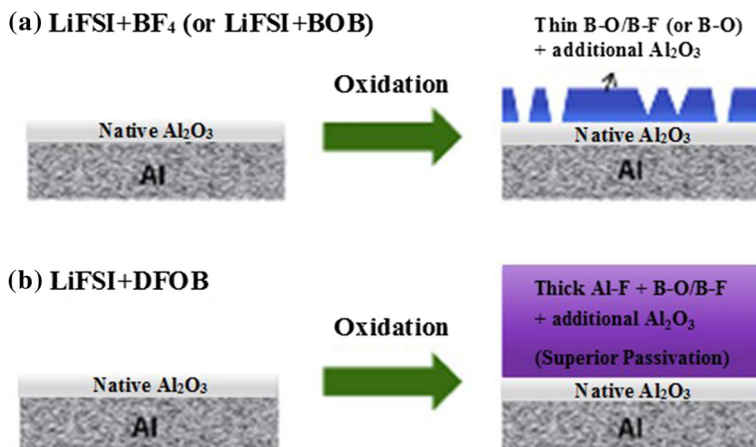


Fig. 17 Schematic diagram of passive layer formation on an Al electrode cycled in EC/DEC (3:7 v/v) electrolyte containing **a** LiFSI + LiBF₄ or LiFSI + LiBOB and **b** LiFSI + LiDFOB salt blends. Redrawn from Ref. [84] Copyright (2015) with permission of Elsevier

A study by Li et al. revealed the effective suppression of Al dissolution in 1 M LiTFSI in EC/EMC (4:6 by wt.) electrolyte by addition of LiDFOB as co-salt (2 wt%), comparable to the pure 1 M LiPF₆-based electrolyte.

The performed XPS measurements revealed the presence of B, F, O, N and S elements in the passivation layer as a result of LiDFOB and LiTFSI decomposition, thus pointing out the presence of B–O species as meritorious for this protection effect [86]. Besides LiDFOB, the presence of LiBF₄ had a positive influence on Al dissolution behavior due to the tendency of passive layer formation [126]. As stated by Zhang et al., the following stability order of salts was established in a EC/DME solvent blend: LiBF₄ > LiPF₆ > LiIm [124]. The Al pitting in LiTFSI-based electrolyte could be depressed by LiBF₄ as co-salt, as shown by Song et al. [127]. By means of electrochemical quartz crystal microbalance (EQCM)—CV and SEM techniques, the authors revealed passivation of the Al current collector to take place as well as suppression of the dissolution process in a 0.5 M LiTFSI + 0.5 M LiBF₄ in EC/DMC (1:1 by wt.) electrolyte. FT-IR spectroscopy and EDX analysis indicated Al[N(SO₂CF₃)₂]_{3-y}(OH)_y (y = 0, 1, 2) as the main dissolution product, RCO₂M (M = Al and/or Li) as organic species and lithium oxalate as well as B-F containing compounds as Al current collector passivation products. In view of these findings, the LiTFSI + LiBF₄ salt blend was assumed to increase the resistance of the Al current collector by passivation [127]. When Al was polarized to high potential values (5.5 V vs. Li/Li⁺), a passivation film, formed on the Al surface, mainly contained Al_xBO₃ species [128]. LiBOB was included in several salt blends, e.g. blends with LiTFSI and LiFSI, due to its Al passivation features [77, 87].

Various approaches were suggested to improve the stability of the Al current collector at high potentials (>3.8 V vs. Li/Li⁺) in LiFSI- and LiTFSI-based electrolytes, for instance by using new solvents [52, 129, 130] or by blending several salts with LiFSI or LiTFSI, e.g. with LiPF₆ [49], LiBF₄ [127] and LiBOB [125].

3.4 Summary

The conducting salt LiPF_6 was identified as the skeletal electrolyte component of SOTA electrolytes; however, its well-known drawbacks limit the performance of the lithium ion cells beyond the temperature range: -20 to 50°C . The blending concept contributes to overcoming the handicaps of the single salt-based electrolyte with the aim to achieve improvements in the transport properties as well as the thermal stability of the nonaqueous electrolyte, enhancement of cycling efficiency of the cell and finally for bringing more safety to consumer electronics. Under the synergistic effect, salt blends combine the advantages of the individual salts and result in improved overall cell performance. In this chapter, the effect of different salt blends on the physicochemical and electrochemical features of nonaqueous electrolytes for LIB was discussed and compared to the single salt formulations. Literature examples of blended salt electrolytes that revealed the influence of blending on anode and cathode performance, suppression or inhibition of Al dissolution as well as the overall cell performance are summarized in Table 8.

Although a salt-blend approach seems as a promising strategy for tuning the nonaqueous electrolytes towards improved performance, there are still important questions left unanswered. Among them, the exact mechanisms of the synergistic effect of blended salts remain unclear as well as whether the beneficial effect on electrochemical performance can be maintained at high and low temperatures or not.

4 Blended Additive Containing Electrolytes

For improvement of SOTA electrolyte formulations, the addition of small amounts of electrolyte additives is a low cost and effective strategy. The specific function of the additives is directed to: (1) improvement of the interfaces/interphases at anode and cathode, (2) stabilization of the lithium salt, (3) improvement of safety such as overcharge protection and flame retardant, (4) improvement of wetting properties, (5) trapping of undesired components and (6) improvement of electrolyte conductivity [131]. In recent years, research expanded from single additives on the electrolyte additive blends for combining different functions in one electrolyte formulation. With this in line, in this chapter, the status of research in the field of blended additive systems is reviewed and comparatively discussed.

4.1 Safety Protection Agents

4.1.1 Flame Retardant Additive Blends

Safety is still a major issue in LIBs due to the high flammability of the nonaqueous organic solvents present in SOTA electrolytes [131, 132]. Any attempt to decrease the flammability of the electrolyte, will pave the way for safer LIBs, for instance by using flame retardant additives [131] or fluorinated compounds as electrolyte solvents [133–136]. Among flame retardant additives, organophosphorous compounds, i.e. trimethylphosphate (TMP) [137] were well studied for the application in

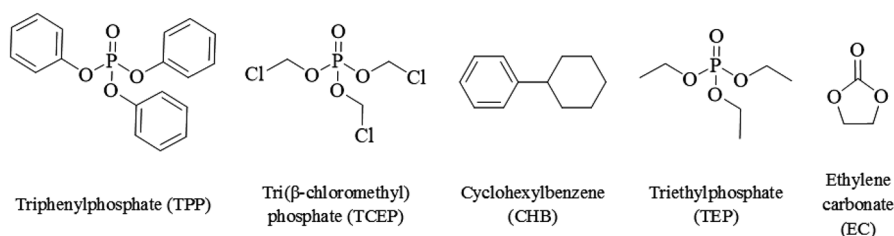
Table 8 Major roles of the discussed lithium salt blends in investigated electrolytes

Lithium salt blend		Target improvement	Investigated electrolyte	References
LiPF ₆	LiBOB	Promoted SEI formation	5 mol% LiBOB in LiPF ₆ -based EC/PC (1:1 v/v)	[107]
			5 mol% LiBOB in LiPF ₆ -based EC/PC/EMC (3:3:4 v/v/v)	
		Suppressed Mn dissolution in LMO	0.5 M LiBOB + 0.5 M LiPF ₆ -based EC/EMC (1:1 v/v)	[72]
		Improved thermal stability of the graphite/LMO		
LiPF ₆	LiBOB, LiFSI	Improved conductivity of the electrolyte for high power applications		
		Improved capacity retention of MCMB/NMC full cells at 55 °C	0.5, 1.0 wt% LiBOB + 1.2 M LiPF ₆ in EC/PC/DMC (1:1:3 by wt.)	[73]
		Promoted SEI formation	0.8 M LiPF ₆ + 0.2 M LiFSI + 0.2 M LiBOB in EC/EMC (3:7 v/v)	[77]
		Stabilization of the Al surface		
LiPF ₆	LiBOB, LiDFOB	Improved cycling performance in graphite/Li and graphite/LFP full cells		
		Achieving high C-rates up to 1.5C with >80% of the initial specific discharge capacity for carbon/NMC full cells	0.65 mol kg ⁻¹ LiPF ₆ + 0.05 mol kg ⁻¹ LiBOB + 0.05 mol kg ⁻¹ LiDFOB in EC/DMSN (80:20 wt.)	[75]
LiPF ₆	LiDFOB	Distinct improvement in cycling efficiency and specific capacity of artificial graphite (AG)/Li half-cells	0.2 M LiPF ₆ + 0.8 M LiDFOB in EC/PC/DMC (1:1:3 v/v/v)	[83]
		Improvement in cycling performance of AG/LFP 063048-type cells at 65 °C		
		Promoted SEI formation	1–5 mol% LiBOB + 1 m LiBF ₄ in PC/EC/EMC (1:1:3)	[95]
		Improved cycling performance in graphite/Li half-cells	1–5 mol% LiBOB in 1 m LiBF ₄ in PC/EC (1:1)	
LiBF ₄	LiBOB	Facilitated SEI formation on Cu surface	0.5 M LiBF ₄ + 0.5 M LiBOB in EC/DEC (3:7 by wt.)	[62]
		Promoted SEI formation	LiBF ₄ + LiBOB blend in PC/EC/EMC (1:1:3 v/v)	[87]
		Improved thermal stability		
		Suppressed Al dissolution		
		Promoted SEI formation	1 M LiBF ₄ + LiBOB (8:2) in PC/EC/EMC (1:1:3 v/v)	[76]
		Wide operating temperature range	1 M LiBF ₄ + LiBOB (8:2) in PC/EC/EMC:MB (1:1:2:1 v/v/v/v)	

Table 8 continued

Lithium salt blend		Target improvement	Investigated electrolyte	References
LiBF ₄	LiTFSI	Suppressed Al dissolution	0.2 M LiBF ₄ + 0.8 M LiTFSI in EC/DMC (1:1) 0.5 M LiBF ₄ + 0.5 M LiTFSI in EC/DMC (1:1)	[127]
LiBF ₄	LiDFOB	Improved capacity and capacity retention of graphite/LNMO full cells at RT and 60 °C Improved cycling ability in high rate cycling at −20 °C Improved ionic conductivity, high thermal stability and wide potential window	0.2 M LiBF ₄ + 0.8 M LiDFOB in EC/DMC/EMC (1:1:1 by wt.) 0.8 M LiBF ₄ + 0.2 M LiDFOB in EC/DMC/EMC (1:1:1 by wt.) 0.5 M LiBF ₄ + 0.5 M LiDFOB in EC/PC/DMC (1:1:3)	[78] [82]
LiBOB	LiBSO ₄ F ₂	Increased anodic stability of the electrolyte for high potential applications Stabilized surface of NMO electrode	0.1 M (LiBSO ₄ F ₂) + 0.7 M LiBOB in EC/DMC (1:1 v/v)	[109]
LiFSI	LiDFOB	Inhibited Al dissolution Comparable rate capability to LiPF ₆ -based electrolyte	0.8 M LiFSI + 0.2 M LiDFOB in EC/DEC (1:1 v/v)	[84]
LiTFSI	LiDFOB	Effective suppression of Al dissolution Improved Coulombic efficiencies Outstanding thermal and water stability, suppression of iron dissolution in LFP-based cells	0.6 M LiTFSI + 0.4 M LiDFOB in EC/EMC (4:6 by wt.)	[86]

LIBs, however a high concentration (>35 wt%) was necessary to suppress the flammability of the electrolyte formulation [137–140], unfortunately leading to higher viscosity, lower ionic conductivity and poor graphitic compatibility. One way to overcome the poor cathodic stability was related to fluorination of organophosphorous compounds. This resulted in a formation of an effective SEI [43, 141–143]. Another advantage of the fluorinated compounds refers to their ability to suppress flammability. The additives investigated in this behalf are displayed in Scheme 4.

**Scheme 4** Molecular structures of the flame retardant additives

Several additive blends combining flame retardant and SEI/CEI formation features were reported by Nam et al. [144] and Shim et al. [145]. These studies investigated the performance of 3 wt% triphenylphosphate (TPP) with different film-forming additives as blends in 1.1 M LiPF₆ EC/EMC (4:6 v/v) electrolyte for graphite/LCO full cells. In an additive blend, TPP had the role of a flame retardant [146–152] by scavenging hydrogen radicals in the flame, thus prohibiting chain reactions and flame propagation [138, 153]. Additionally, it formed a fire resisting char layer that suffocated the flame but the flash point was not significantly altered. This finding indicated decomposition of TPP essential for the action as a flame retardant [146]. The thermal stability was investigated by means of DSC measurements. Nam et al. [144] and Shim et al. [145] showed that in case of the TPP containing blended electrolyte formulation, the endothermic peak could be shifted to higher temperature by 20 °C compared to the standard electrolyte. The blend of TPP and vinyl acetate (VA) showed even higher thermal stability since the endothermic peak could be shifted to higher temperature by 25 °C. Another example of a flame retardant and film-forming additive blends was reported by He et al. [154]. Their study focused on the additive blend tri(β-chloromethyl) phosphate (TCEP) together with cyclohexylbenzene (CHB) in 1 M LiPF₆ EC/DMC/EMC (1:1:1 by wt.) electrolyte. TCEP is known as a flame retardant [154], whereas CHB is an additive that protects the cell during overcharge [155]. He et al. showed that TCEP absorbs hydrogen radicals during the decomposition and therefore reduces the possibility of endothermic chain reaction induced by free radicals. As shown in Fig. 18, the addition of TCEP to the electrolyte reduced the flammability, whereas the addition of CHB led to an increase. The blend of 5 wt% TCEP and 5 wt% CHB showed the lowest flammability, indicating a high impact of TCEP on the SET and therefore improved safety characteristic of the investigated electrolyte.

So far, there has always been a trade-off between achieving non-flammability and, at the same, time decreasing cell performance (faster capacity fading/lower

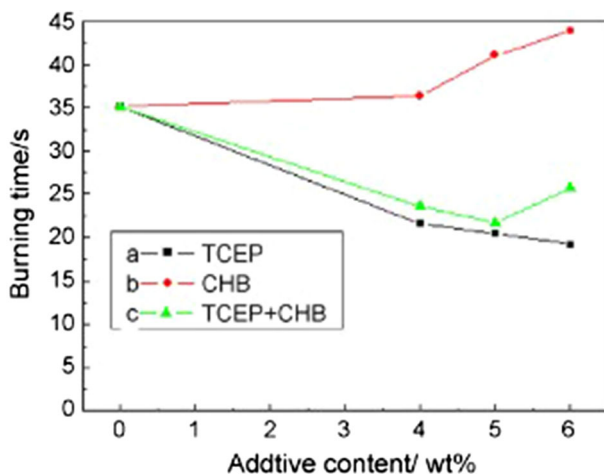
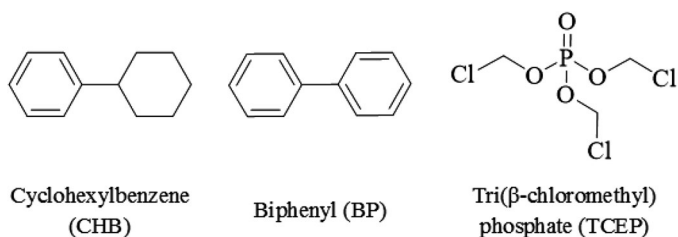


Fig. 18 SET determination of the 1 M LiPF₆ EC/DMC/EMC (1:1:1 by wt.) electrolyte with TCEP and CHB as additives. Reprinted from Ref. [154]. Copyright (2007) with permission of Elsevier

power) or deteriorating physical influences (high melting point/high viscosity) by using flame retardants in combination with organic solvents [138, 156]. Therefore, RTILs are considered as one of the safest electrolyte components since they are flammable only at very high temperatures, non-volatile, and relatively thermally stable [157, 158], depending on the respective IL composition. Furthermore, they have a wide liquidus range and relatively wide ESW, in particular a high oxidation stability of ILs with fluorinated anions [159, 160]. However, IL-based LIB electrolytes are often highly viscous and thus exhibit low ionic conductivity compared to conventional nonaqueous electrolytes [22, 161]. Using a small amount of organic compound as an additive is an alternative way to overcome aforementioned limitations. On the one hand, these additives will improve the conductivity by reducing the viscosity. On the other hand, the addition of organic solvents will significantly affect the flammability and thermal properties of the electrolyte. For this reason, a non-flammable additive is required to guarantee safe operation of the cell. Lalia et al. [162] investigated *N*-methyl-*N*-propylpiperidinium bis(trifluoromethylsulfonyl)imide (PP₁₃-TFSI) with addition of triethylphosphate (TEP) and EC in a volume ratio of 80:10:10 in graphite/LMO full cells. The alkylphosphate TEP showed low viscosity, low melting point, high boiling point, non-flammability and a relatively high dielectric constant [137]. TEP as an additive was added to suppress the flammability of the organic electrolyte, to improve the ionic conductivity, viscosity and thermal stability of the PP₁₃-TFSI-based electrolyte. The disadvantage of this compound was related to the instability towards reductive decomposition on the graphite anode surface and a poor long-term cycling stability [162]. EC, commonly used as electrolyte solvent [163], whereas here in additive amounts, had the role to form an effective SEI between a graphite-based negative electrode and the nonaqueous electrolyte. The direct flame test of TEP and EC binary blend in 0.4 M LiTFSI/PP₁₃-TFSI electrolyte showed that with an equal ratio of TEP and EC, the system was completely non-flammable and could thus serve as a safe electrolyte for LIBs. Furthermore, TEP with a high dielectric constant and low viscosity was able to increase the ionic conductivity and to improve the electrochemical performance, as well.

4.1.2 Overcharge Protection Additive Blends

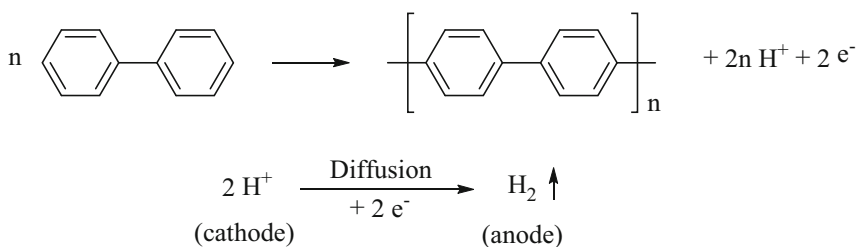
Unlike aqueous electrolyte batteries, where the decomposition of the solvent water is the basis for ‘the oxygen-cycle’ [164, 165], the nonaqueous (water-free) electrolyte LIB technology has no intrinsic chemical/electrochemical overcharge protection included in the battery solvent [166]. Overcharge can result in severe performance degradation and safety problems [167]. Overcharge protection additives can be categorized as reversible redox shuttle- and irreversible shut down additives [131]. One promising class of the latter category are compounds that polymerize electrochemically on the surface of the cathode or create, by decomposition, reactive species that induce the polymerization of the base electrolyte [155, 168–170]. Vinylene [171–174] and isocyanate [175] compounds are known to electrochemically polymerize both during reduction [176] or oxidation [177]. The over-charge additives discussed in this behalf are displayed in Scheme 5.



Scheme 5 Molecular structures of the overcharge protection additives

Biphenyl (BP), for instance, was studied as an aromatic shutdown additive [144, 168, 178]. The mechanism can be described as follows: (1) BP is oxidized at the cathode side, (2) the oxidized products form through radical coupling a poly(*p*-phenylene)-type polymer that is depositing on the cathode surface and growing through the separator to anode side, creating an electronic short circuit that carries a part of the over-charge current, (3) the co-generated protons diffuse through the electrolyte and to the anode side, (4) at the anode are reduced to hydrogen increasing the internal and cell pressure and (5) the pressure increase in the cell is activating a current interrupt device (CID) that is interrupting the external current circuit, this the current causing over-charge is interrupted (Scheme 6) [168, 179].

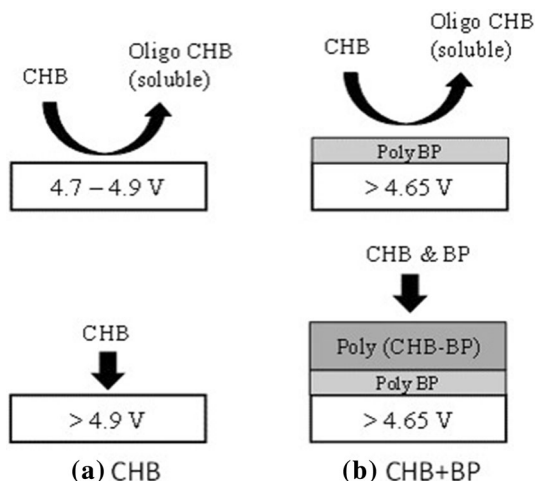
The oxidation products can be described as condensed aromatic compounds with a widespread π -electron conjugation system and the bond formation mainly occurs at the *ortho*-positions of the phenyl group [178]. However, a high concentration of BP within the electrolyte would lead to a decrease in lithium cycling efficiency. The hydrogenated form of BP is CHB and has a higher oxidation potential than BP, making it more inert towards undesirable side reactions during operation in the normal voltage range of the cell. CHB also shows higher lithium metal cycling efficiency than BP, allowing the interpretation that it is more stable towards charged anode electrodes [155]. BP and CHB both belong to overcharge protection agents called non-redox shuttle type, with the same overcharge protection mechanism during electrochemical oxidation [178]. Lee et al. found that the co-use of CHB and BP in 1 M LiPF₆ EC/EMC (1:2 v/v) in graphite/LCO full cells is much more effective for overcharge protection than using BP or CHB alone (Table 9) [180]. The synergistic effect in overcharge protection was proven in pouch cells with a nominal capacity of 760 mAh by applying a constant current with a voltage limit of



Scheme 6 Proposed mechanism of electropolymerization and hydrogen evolution of BP

Table 9 Number of the passes over total trials during overcharge tests for graphite/LCO full cells with various CHB and BP composition in 1 M LiPF₆ EC/EMC (1:2 v/v) Adapted from [180]

CHB (%)	BP (%)	12 V and 1 A passes/trials	12 V and 2 A passes/trials
0.0	0.0	0/3	0/3
3.0	0.0	0/1	1/2
4.0	0.0	2/2	8/10
3.5	0.5	–	10/10
3.0	1.0	3/3	3/3
3.0	2.0	3/3	3/3

Fig. 19 Electropolymerization process for **a** CHB and **b** CHB + PB on Pt electrodes. Redrawn from Ref. [180]

12 V and a magnitude of the current among 0.5, 1 and 2 A. Prior to the overcharge test, cells were cycled to 4.2 V several times and were fully discharged in the last cycle. It was shown that the addition of CHB helps to increase the safety in case of overcharge but the electrolyte containing blend with BP is much more effective than the CHB containing electrolyte. The blend of CHB and BP expanded the safety region up to 12 V and 2 A, which the single additive CHB could never reach. This synergistic effect was studied in electropolymerization measurements on Pt electrodes and explained as following (Fig. 19): CHB alone formed a polymer film on the Pt surface above 4.9 V. At lower values, the oxidation products of CHB seem to be too soluble to precipitate or the adhesion to the Pt surface is too weak. By increasing the concentration of the oxidation products, a polymer film can be formed on the surface.

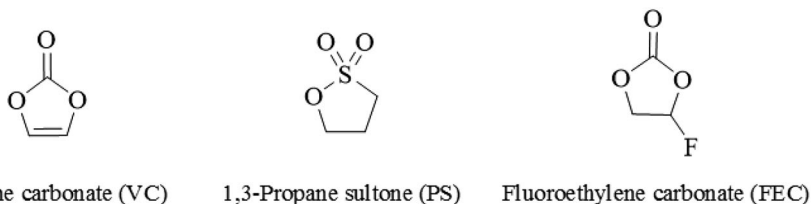
In contrast, when using the blend, BP formed a polymer film from 4.65 V. The formed poly-BP film, resulting in much more effective polymer deposition on the surface, catalyzes the oxidation of CHB and probably BP. The application of a shutdown and flame retardant additive at the same time was disclosed by He et al. [154]. Here, the electrolyte blend contained 5 wt% CHB as shutdown and 5 wt%

TCEP as flame retardant additive to improve the safety characteristics of the investigated cells. Hereby, the surface analysis of LiCoO_2 electrodes after overcharge indicates a polymer film on the surface, which can be attributed to CHB. Furthermore, the temperature change during overcharge showed an exothermic peak after 30 min, which proves the exothermic polymerization process of CHB. This generated heat can be absorbed by chloroethane, a decomposition product of TCEP. Through this synergistic effect, the increase of temperature can be delayed.

4.2 SEI and CEI Film-Forming Additive Blends

Utilization of electrolyte additives that form in situ SEI on the anode/CEI on the cathode side is one of the most effective methods to suppress further electrolyte decomposition and protect the electrode active material from degradation, thus decreasing the internal resistance of the cell [131, 181, 182]. A blend of different film-forming additives can be used either to form an effective SEI and CEI at the same time or to strengthen the surface film-formed by one single additive. Small amounts of organic solvents as additives were used to form an electrochemically effective SEI prior to reduction of IL [131, 183, 184]. With this in line, Kim et al. investigated a binary additive blend based on 1-butyl-1-methylpyrrolidinium bis(trifluoromethylsulfonyl)imide (BMP-TFSI) with 10% (1:1 by wt.) VC and 1,3-propane sultone (PS) (Scheme 7) in LTO/LCO full cells [185]. VC is a well-known and intensively studied SEI additive [143, 174, 186–188]. It improves thermal stability of the surface layer due to the formation of polymeric species [189], well known also for other vinylene compounds [138–141]. Furthermore, PS is able to form an SEI on the anode [184, 190–192]. Xu et al. [184] reported PS containing electrolyte formulations to modify the cathode surface, which resulted in an improved cycling performance. The addition of organic solvents suppresses the reductive decomposition of BMP-TFSI. Moreover, high discharge capacities and improved capacity retention could be achieved by using the combination of VC + PS. FTIR measurements of the blended system showed that the film on the LTO electrode surface consists of a polymeric/organic and inorganic salt species blend and was more compact and dense when using the additive blend instead of VC or PS alone [185].

Additionally, VC is a well-known and studied SEI additive for silicon (Si)-based electrodes [193, 194]. This alternative anode material displayed a high theoretical Li storage capacity, volumetric capacity as well as gravimetric capacity [195, 196].



Scheme 7 Molecular structures of the film-forming additives

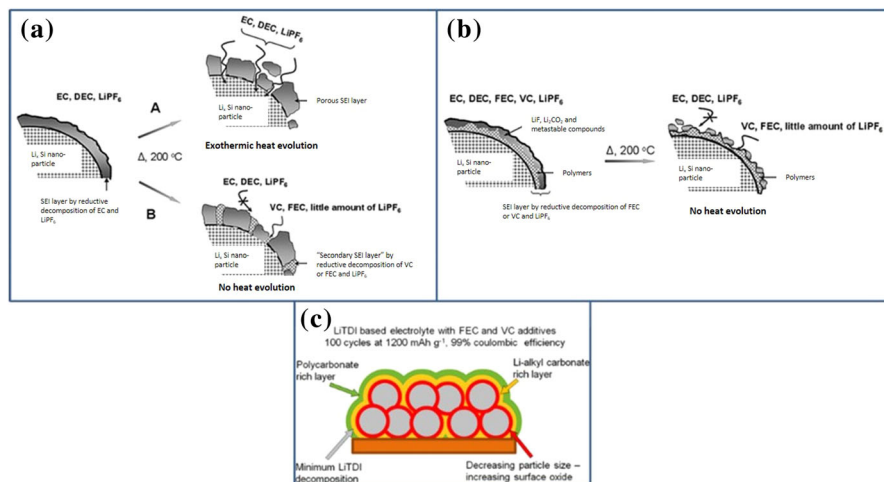
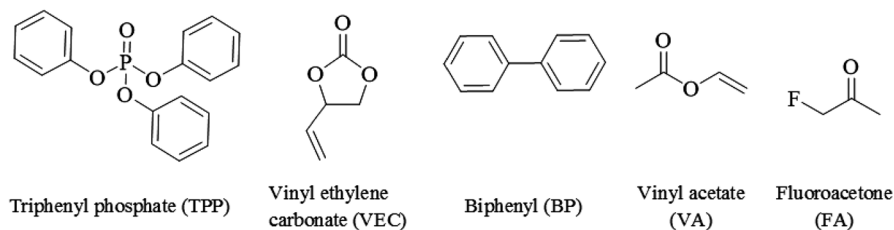


Fig. 20 **a** A schematic diagram showing the thermal events in the Li_xSi-Ref system that occur while heating up to 200 °C (route A). Thermal degradation of the SEI layer makes it porous and permeable to the electrolyte components, such as EC, DEC, and LiPF₆. Direct contact between the components and Li_xSi initiates the thermal runaway. The thermal reactions of Li_xSi that was lithiated in the Ref electrolyte, washed, dried and heated in contact with either FEC10 or VC10 are shown in route B. Upon thermal degradation, the FEC and VC additives in contact with Li_xSi provide the “secondary SEI” layer, which does not allow direct interaction between Li_xSi and the electrolyte. **b** A schematic diagram showing the protective function of the FEC- and VC-derived SEI layers on the surface of the lithiated Si nano-particles during heating up to 200 °C. The SEI layer, which forms electrochemically, is mostly composed of a polymeric species and shows excellent stability at elevated temperatures. **c** A schematic drawing of the formed SEI in the presence of both additives FEC and VC. The inner shell mainly consists out of lithium alkyl carbonates (yellow), whereas the outer shell mainly consists out of polycarbonates (green). Redrawn from Ref [143, 202]. Copyright (2013 and 2016) with permission of Elsevier and ACS

However, large volume expansion upon extraction and insertion of Li led to a decreased electronic conductivity resulting in capacity fading [197]. Besides VC, fluoroethylene carbonate (FEC) (Scheme 7) is another thoroughly studied SEI additive for Si-based electrodes [198–201]. The thermal stability of VC and FEC was studied by Profatilova et al. [143]. The authors showed that the SEI formed by the 1 M LiPF₆ in EC:DEC (3:7 by wt.) electrolyte cracked open by heating up to 200 °C, thus resulting in a formation of a permeable SEI layer. This allowed the electrolyte to interact with the Li_xSi nano-particles. When using the electrolyte containing either 10 wt% VC or FEC, a so-called secondary SEI layer could be formed. The decomposition reactions driven by VC or FEC were similar and resulted in formation of polycarbonates, thus preventing direct contact between the electrolyte formulation and Li_xSi nano-particles. It was induced immediately when the SEI cracked open at elevated temperatures (Fig. 20a). Furthermore, VC and FEC formed a so-called primary SEI layer during charge, which was mainly formed by polycarbonate species. The decomposition layer of 1 M LiPF₆ in EC:DEC (3:7 by wt.) was formed on top of it. By heating up this electrode to 200 °C only the outer shell of the primary SEI cracked open. Nevertheless, the Li_xSi nano-particles were still protected by the polymers formed by either VC or FEC, preventing the contact between the electrode and electrolyte (Fig. 20b).

The combination of both SEI additives for the application of Si anode materials was first studied by Mazouzi et al. [203]. The investigated electrolyte was 1 M LiPF_6 in either EC/DMC (1:1 by wt.) or EC/DEC (1:1 by wt.) with 10 wt% FEC and 2 wt% VC in Si/Li half-cells. They showed that using this additive blended electrolyte formulation, the kinetics of the electrolyte degradation was slowed down due to the prior reduction of VC and FEC compared to EC. The reduction products were more stable and limited the EC reduction, thus resulting in a strongly improved cycling performance. The work of Gauthier et al. [204] confirmed those results. They used 1 M LiPF_6 in EC/DEC (1:1 by wt.) with 10 wt% FEC and 2 wt% VC as a blended electrolyte formulation in Si/Li half-cells. The cells could be charged up to 900 cycles with a Coulombic efficiency of 99%. The synergistic effect of both SEI additives was studied by Lindgren et al. in Si/Li half-cells containing 0.6 M 4,5-dicyano-2-(trifluoromethyl) imidazolidine (LiTDI) in EC/DMC (2:1 v/v) with 5 vol% FEC and 1 vol% VC [202]. Besides the improved cycling performance up to 100 cycles with a Coulombic efficiency of 99%, they showed the importance of LiTDI and the blended electrolyte formulation. The additive blend significantly contributed the SEI formation; otherwise, an ineffective SEI was formed and accelerated lithium silicate formation. LiTDI was only partially involved in the SEI formation, whereas the main part was LiF, formed by decomposition of FEC. The formed SEI consisted of an inner and outer shell. The inner shell mainly consisted of the decomposition products driven by the lithium alkyl carbonates, whereas the outer shell mainly of polycarbonates (Fig. 20c). They claimed that the high content of LiF as well as the large extent of polycarbonates was essential for the cycling performance of Si anodes. Furthermore, as mentioned above, Nam et al. [144] and Shim et al. [145] studied blended additive electrolytes containing the flame retardant TPP together with different film-forming additives such as vinyl ethylene carbonate (VEC), biphenyl (BP), VA and VC. Their molecular structures are displayed in Scheme 8.

VEC could be reduced at potentials above 1 V vs. Li/Li^+ due to the electron-rich double bond and was therefore found to be a good SEI-film-forming additive [127, 205–211]. During reduction on the anode, VEC polymerizes with the major film components Li_2CO_3 and ROCO_2Li . When the concentration of VEC is low, ROCO_2Li formation will be small. This results in an improved electrochemical performance, whereas a high content of VEC suppresses the formation of Li_2CO_3 causing an adverse effect on the cycling performance [208]. Additionally, VEC is able to form a CEI layer even at elevated temperatures [206, 212, 213]. BP is able to



Scheme 8 Molecular structures of the film-forming additives

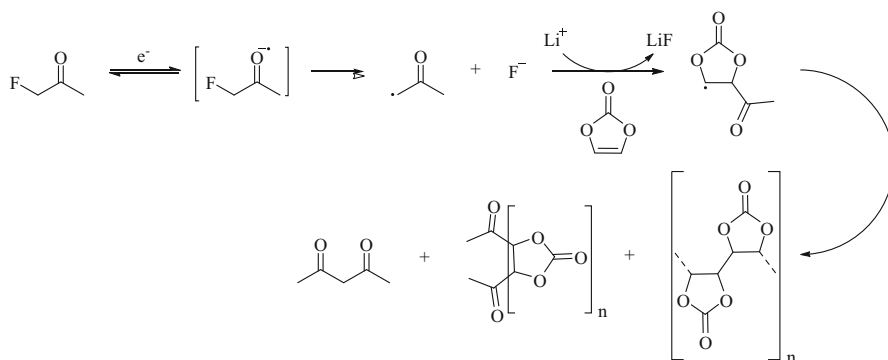
form a thin CEI film when only present in small amounts (0.1 wt%), which prevents further electrolyte decomposition on the cathode side [214]. However, the use of BP had several drawbacks, e.g. poor lithium cycling efficiency due to the reaction with lithium metal and no oxidation stability during prolonged storage or cycling [168, 169]. VA is known to form a SEI layer on the anode surfaces, resulting in an enhancement of cycling performance [215, 216]. This film-formation was due to the reduction of the vinyl group induced by electro-polymerization [171–173, 216]. As previously stated, TPP was used as a flame retardant additive to increase the thermal stability. However, in an amount of up to 10 vol%, TPP influenced the surface chemistry on both electrodes, whereas the cycling performance and capacity retention were not affected [146, 148]. On the cathode side, high additive concentration gave rise to the formation of a thick protective layer. On the anode side, TPP did not go through electrochemical reduction but it participated in the SEI layer formation by chemical reaction [147, 217]. Furthermore, the cell impedance increased which affects the cycling performance and the electrochemical stability up to 5 V vs. Li/Li⁺ [148, 150, 152, 217]. Nam et al. [144] and Shim et al. [145] found that all blended additive electrolyte formulations of 3 wt% TPP with 1 wt% VEC, VA, VC or 0.1% BP in 1.1 M LiPF₆ in EC/EMC (4:6 v/v) showed an improved cell performance compared to the single additive electrolyte formulations containing TPP. Furthermore, the electrochemical stability, the irreversible capacity, the Coulombic efficiency and the rate capability were improved, as listed in Table 11. The TPP + BP containing electrolyte showed better cycling stability than the electrolyte with TPP + VEC blend. Moreover, the electrolyte formulation containing TPP + BP was able to maximize the discharge capacity and to reduce the cell impedance. This showed that BP can improve the cell performance and thermal stability better than VEC [144]. By comparing the effect of VA or VC in the binary electrolyte formulation, the VC containing formulation showed the best enhancement in thermal stability and cell performance [145]. In addition, Shim et al. [218] investigated the ternary blend of TPP with VA + VC and the blend of TPP with VEC + BP. Both additive blended electrolytes displayed high electrochemical stability up to 4.9 V vs. Li/Li⁺. The impedance of a cell containing TPP + VEC + BP could be reduced during cycling and therefore the specific discharge capacity was maximized. Furthermore, the cycle life performance was better for the TPP + VEC + BP blend than for the TPP + VA + VC blend. With this in line, Shim et al. [218] reported TPP + VEC + BP additive blend in organic carbonate-based electrolyte formulation to contribute to the improved cell performance and thermal stability (Table 10).

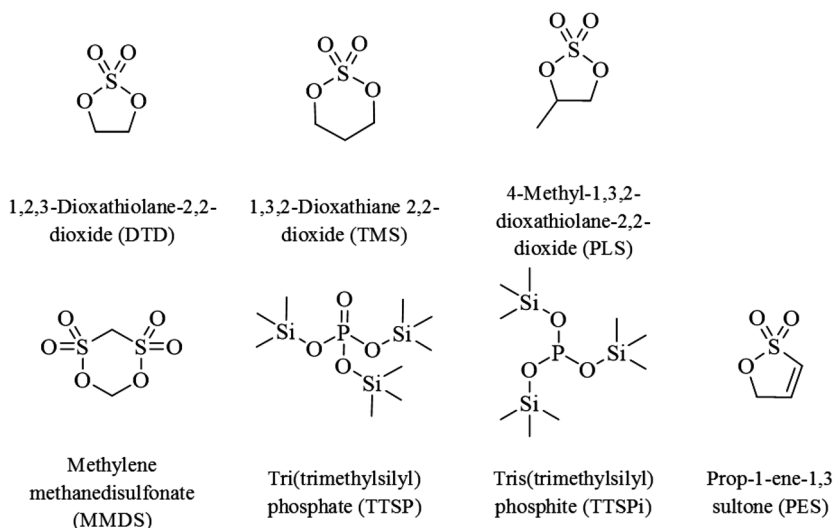
VC, known as one of the most common SEI forming additives, could only be used in concentrations up to 2% as higher concentrations led to a larger charge transfer resistance, especially at the graphite electrode [219]. For this reason, other SEI forming additives were investigated in combination with VC. For instance, Krämer et al. [220, 221] investigated 1 wt% VC with 1 wt% fluoroacetone (FA) in 1 M LiPF₆ in PC electrolyte using graphite/Li half-cells. PC-based electrolyte formulations showed a poor comparability with graphite electrodes, because the PC molecules co-intercalated into the graphite structure thus resulting in exfoliation [187]. FA was able to suppress this phenomenon by forming a protective SEI [220].

Table 10 Comparison of the effects of the addition of one or two film-forming additives (VEC, BP, VA, VC) to the electrolyte formulation containing 3 wt% TPP in terms of irreversible capacity, Coulombic efficiency and capacity retention

Electrolyte formulation	Irreversible capacity	Coulombic efficiency (%)	Capacity retention (40 cycles) (%)
1.1 M LiPF ₆ EC/EMC (4:6 v/v)	2.5 mAh g ⁻¹ [144], 0.39 mAh [145]	98.1 [144], 89.0 [145]	68 [144], 67 [145, 218]
1.1 M LiPF ₆ EC/EMC (4:6 v/v) + 3% TPP	3.3 mAh/g [144], 0.83 mAh [145]	97.4 [144], 77.0 [145]	65 [144], 64 [145, 218]
1.1 M LiPF ₆ EC/EMC (4:6 v/v) + 3% TPP + 1% VEC	2.1 mAh g ⁻¹ [144]	98.4 [144]	70 [144]
1.1 M LiPF ₆ EC/EMC (4:6 v/v) + 3% TPP + 0.1% BP	0.8 mAh g ⁻¹ [144]	99.4 [144]	75 [144]
1.1 M LiPF ₆ EC/EMC (4:6 v/v) + 3% TPP + 1% VA	0.72 mAh [145]	81.0 [145]	70 [145]
1.1 M LiPF ₆ EC/EMC (4:6 v/v) + 3% TPP + 1% VC	0.31 mAh [145]	91.0 [145]	68 [145]
1.1 M LiPF ₆ EC/EMC (4:6 v/v) + 3% TPP + 1% VA + 1% VC	–	–	69 [218]
1.1 M LiPF ₆ EC/EMC (4:6 v/v) + 3% TPP + 1% VEC + 0.1% BP	–	–	71 [218]

Reactive species like radical anions, formed during reduction, were able to eliminate fluorine anions. This resulted in an increase of the LiF amount in the formed SEI [221]. The combination of two different SEI additives in 1 M LiPF₆ in PC led to stable cycling performance in graphite/Li half-cells. Furthermore, a higher capacity retention, a higher first cycle charge/discharge efficiency and a good discharge rate could be achieved [220]. The formed acetone and VC radical species reacted with each other and formed copolymers responsible for the aforementioned improvements [221]. The postulated mechanism is illustrated in Scheme 9.

**Scheme 9** Postulated reductive decomposition mechanism of FA in the presence of VC. Redrawn from [221]



Scheme 10 Molecular structures of the film-forming additives

Another possibility to improve the SEI formation by VC was to add sulfur-containing additives, known for SEI formation on the negative electrode [133, 174, 222–224]. Li et al. [223] and Ota et al. [222] suggested that the formed SEI in presence of sulfur containing additives contained both inorganic materials like Li_2SO_3 and organic compounds like ROSO_2Li . These organic sulfur-containing additives reacted at the graphite surface in such a way that sulfur was incorporated into the SEI film. Zhang et al. [225] and Han et al. [226] studied the combination of 1.5 wt% VC and PS in 1 M LiPF_6 in EC/EMC (3:7 by wt.) in graphite/NMC and graphite/Li half-cells (Scheme 10). This additive blend led to a SEI formation that combines the advantages of both electrolyte additives resulting in an improved cycling performance: (1) PS was responsible for the formation of lithium alkylsulfonate which gave rise to higher conductivity and (2) VC led to the formation of poly(VC) resulting in an improved stability of the SEI. Additionally, Xia et al. [219, 227] studied the effect of the homologous series of cyclic sulfate additives including ethylene sulfate (or 1,3,2-dioxathiolane-2,2-dioxide) (DTD), trimethylene sulfate (or 1,3,2-dioxathiane 2,2-dioxide) (TMS), propylene sulfate (or 4-methyl-1,3,2-dioxathiolane-2,2-dioxide) (PLS) and PS either as a single additive or in combination with VC in 1 M LiPF_6 EC/EMC (3:7 by wt.) electrolyte (Scheme 10). As DTD, TMS and PLS have the same SO_4 unit bonded to a different position in hydrocarbon ring, the impact of the ring size could be studied. DTD has a low LUMO energy and could therefore be easily reduced forming a SEI film which prevented further electrolyte decomposition [228]. Sano et al. suggested that the SEI formed by DTD is due to one electron reduction of the sulfur atom to form a DTD anion radical which could react further with another DTD molecule to form a PEO like polymer and Li_2SO_3 [229]. Furthermore, it was possible to use DTD as a co-solvent for a PC containing electrolyte formulation to fully prevent solvent co-

intercalation and graphite exfoliation [230] and to reduce the volume expansion due to gas formation [228], which was reported to happen in PC electrolytes without the additive [231, 232]. No positive influence on the cycling performance of the graphite could be achieved in the presence of TMS as a single additive, due to the formation of a small amount of sulfate compounds at the electrode surfaces [233]. For the PLS additive containing electrolyte formulation, no improvement compared to the VC containing electrolyte blend could be observed [219]. The results of the blended additive-containing electrolytes were compared with the results from other groups on methylene methanedisulfonate (MMDS) (Scheme 10). MMDS containing electrolyte is able to form a thinner CEI film on the cathode surface, as compared to the reference electrolyte. This resulted in a reduction of the charge transfer impedance and in a high Li^+ ion mobility [234, 235]. Amounting to 1–3 wt%, the additive components were added into the LiPF_6 -based EC/EMC (3:7 by wt.) electrolyte and investigated in graphite/NMC pouch cells. Xia et al. [219] showed that DTD can compete with VC as a single additive, however large amount of gas generation during the formation had to be considered. PLS was not effective neither alone nor in combination with VC and the blend of VC with either DTD, TMS or MMDS led to a better performance. The TMS + VC additive blend containing electrolyte did not gave rise to a significant amount of gas generation during the formation step of graphite/NMC full cell, balanced to operate at 4.2 V. Figure 21 depicts the direct comparison between the three different blended electrolyte formulations in the Coulombic inefficiency, the charge end point capacity slippage, the charge transfer resistance and the voltage drop. Furthermore, Xia et al. demonstrated that the synergistic effect could be either beneficial or detrimental in the additive combination of either PS + VC or MMDS + VC [227]. The electrolyte formulation containing MMDS + VC blend was able to improve

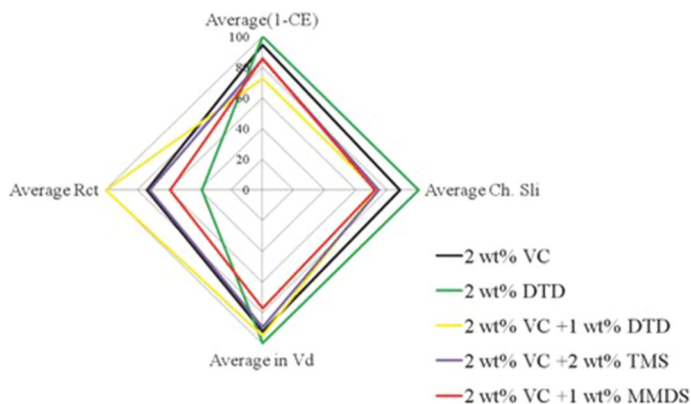


Fig. 21 A “spider” plot that compares the best blends of the additive VC with DTD, TMS and MMDS. The four axes represent the Coulombic inefficiency (1-CE), the charge end point capacity slippage (Ch. Sli), the charge transfer resistance after cycling (R_{CT}) and the voltage drop during storage (V_d). The axes have been scaled so that 100% is the value of the additive that has the largest value of each parameter. The best additive blend would have values closest to the center of the plot. Reprinted from Ref. [219]. Copyright (2014) with permission of ECS

the cycling performance and decreased the swelling of the cell compared to the single additive containing electrolyte formulation. In the case of the of PS + VC additive blended electrolyte, no improvement compared to the single additive containing electrolyte formulation could be achieved in respect to the cycling performance. Furthermore, the PS + VC blend-containing electrolyte showed a drastic increase in the cell impedance, whereas the electrolyte formulation with MMDS + VC additive blend resulted in a decrease compared to the single additive electrolyte formulation. In case of the ternary VC + PS + MMDS blend-containing electrolyte, a decrease of the cell impedance compared to the binary PS + VC blended electrolyte could be observed, but compared to the binary MMDS + VC blend-containing electrolyte, the cell impedance was increased. It could be concluded that the additive PS has a detrimental influence on electrochemistry in combination with VC and MMDS.

The work performed by Sinha et al. focused on organophosphorous compounds, containing a tris(trimethylsilyl) (TMS) group, as additives regarding their possible application as high-voltage additives [236]. TMS-based additives found multiple considerations in research [237, 238]. It was suggested that tris(trimethylsilyl) phosphate (TTSP) and tris(trimethylsilyl) phosphite (TTSPi) as additives could help to improve the cycling performance of lithium ion cells at high voltage and/or elevated temperature (Scheme 10) [239].

TTSP was reported to be able to replace EC in the Li^+ solvation shell and to form a CEI film without increasing the acidity at the cathode interface; acidity, which would be the result of electrolyte solvent oxidation. This effect can be explained by a proton transfer to another TTSP molecule or an intramolecular oxidation instead of a proton transfer to the electrode surface. The presence of this CEI was the reason behind the significant improvement in cycling performance, the decrease in cell impedance and the reduction of transition metal dissolution [240–244]. TTSPi is involved in formation of the CEI film on the cathode surface [245–249] due to the lower oxidation and reduction potential compared to EC [248], resulting in preferential oxidation [249]. Sinha et al. compared the performance of 1, 2 or 3 wt% TTSP or TTSPi together with 2 wt% VC as a binary additive blended 1 M LiPF_6 EC/EMC (3:7 by wt.) electrolyte in graphite/NMC pouch cells [236]. They showed that TTSP or TTSPi do reduce the overall cell impedance. However, in the binary blends of either VC + TTSP or VC + TTSPi, the cell impedance was increased. The best-blended system among investigated, consisted of 2 wt% VC and 1 wt% TTSPi and could lead to a longer cycle life, longer calendar life and reduced impedance compared to cells with electrolytes containing only VC. Ma et al. [250] and Wang et al. [251] combined sulfur and phosphorus containing film-forming additives (mentioned above) with organophosphorus compounds and studied their impact on the cycling performance as a ternary or quaternary systems. As listed in Table 11, all investigated electrolyte blends showed higher Coulombic efficiencies and lower charge end-point capacity slippage rates compared to electrolyte formulations containing VC as a single additive, thus pointing out a reduction in parasitic reaction rates. Furthermore, the self-heating rate was decreased indicating a thermally stable SEI on the anode. A more effective passivation layer on both electrodes was formed, represented by lower charge transfer impedance after

Table 11 Summary of the electrolyte additive combinations discussed in the text and their amount in the electrolyte formulation 1 M LiPF₆ EC/EMC (3:7 by wt.) in graphite/NMC full cells

VC (%)	TMS (%)	DTD (%)	MMDS (%)	TTSP (%)	TTSPi (%)
2.0	1.0	–	–	0.5	0.5
2.0	1.0	–	–	–	–
2.0	–	1.0	–	0.5	0.5
2.0	–	1.0	–	–	1.0
2.0	–	–	1.0	0.5	0.5
2.0	–	–	1.0	–	1.0

cycling compared to the single additive formulation. Additionally, the gas generation during cycling was the same or less as compared to the VC containing electrolyte formulation. All investigated blended electrolyte formulations led to improved safety and longer cycle life with better power capability of the resulting LIBs.

The additives MMDS (1 wt%) and TTSPi (1 wt%) were also investigated together with 2 wt% prop-1-ene-1,3 sultone (PES) in 1 M LiPF₆ EC/EMC (3:7 by wt.) using graphite/NMC full cells by Ma et al. [252, 253], Nie et al. [254] and Madec et al. [255] (Scheme 10). These additive blended electrolyte formulations improved the capacity retention, reduced the cell impedance as well as suppressed gas formation significantly. PES has a similar structure compared to PS, but has a more positive reduction potential, which made the resulting SEI more effective [256, 257]. However, Madec et al. also reported that blend of 2 wt% PES + 1 wt% MMDS + 1 wt% TTSPi displayed a capacity loss and an impedance increase when the cell was charged for a long time at potentials above 4.4 V vs. Li/Li⁺ due to the relatively poor stability of the sulfite species [255]. XPS studies showed that the preferential reaction of MMDS and PES was responsible for more effective SEI and CEI formation. Therefore, a better capacity retention could be achieved, accompanied by reduced gas formation. In addition, they found that TTSPi reacted at the graphite surface before the cell was fully charged. Furthermore, Ma et al. compared the impact of VC or PES in the combination with MMDS and TTSPi in 1 M LiPF₆ EC/EMC (3:7 by wt.) as a blended electrolyte formulation [258]. The PES containing blended additive electrolyte formulations showed a better capacity retention as compared to that of the VC containing one (Fig. 22). The cells containing the additive blend 2 wt% PES + 1 wt% MMDS + 1 wt% TTSPi showed the highest capacity retention which is more than 80% of the initial capacity after 900 cycles at 55 °C. The thermal reactivity for the charged graphite electrode was lower for the PES + VC containing blends, which resulted in a significant life improvement of these LIBs. This was investigated by means of an ARC experiment where 140 mg lithiated graphite reacted with 140 mg of the electrolyte formulation containing 2 wt% of the selected additive. For both additive blended electrolyte formulations, the exothermic feature was significantly decreased.

Burns et al. investigated a binary additive blend containing VC in combination with trimethoxyboroxine (TMOBX) in 1 M LiPF₆ EC/EMC (3:7 by wt.)

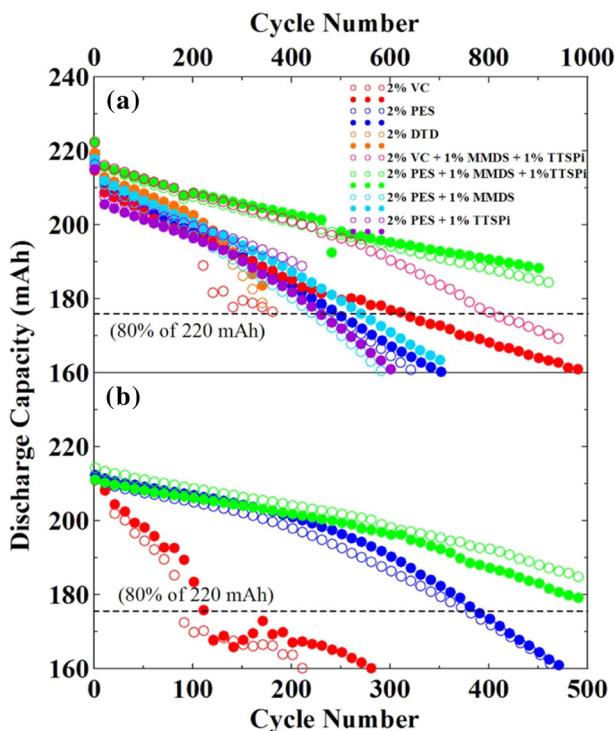
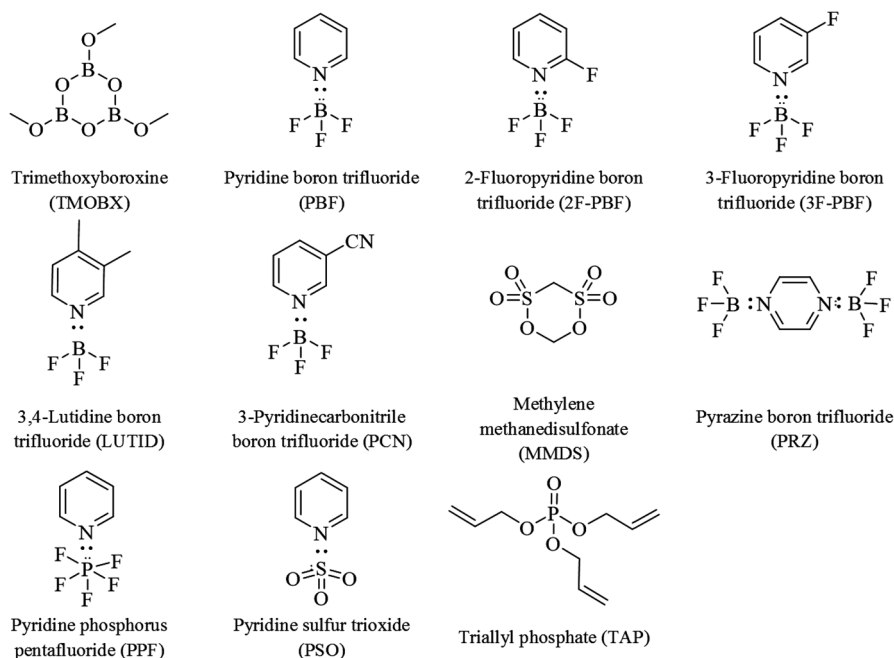


Fig. 22 **a** Capacity versus cycle number for graphite/NMC111 pouch cells (unclamped) containing selected additives or additive blends in the range from 2.8 and 4.2 V, at 55 °C and 80 mA. **b** Capacity versus cycle number for graphite/NMC442 pouch cells (unclamped) containing selected additives or additive blends. The cycling was performed between 3.0 and 4.4 V at 45 °C and at 100 mA. Reprinted from Ref. [258]. Copyright (2015) with permission of ECS

(Scheme 11) [259]. Mao et al. showed that the presence of this methoxy-group in TMOBX helps to reduce the capacity loss during cycling more effectively than the (BO)₃ ring alone [260]. Furthermore, TMOBX in amounts of less than 1% is a very useful additive to decrease the cell impedance [261]. Graphite/Li half-cell investigations showed that the presence of TMOBX increases the rate of parasitic reactions. The first cycle irreversible capacity was increased and no effective SEI layer could be formed [262]. However, Burns et al. [259] and Petibon et al. [263] pointed out that 2 wt% VC together with 0.3 wt% TMOBX reduced the cell impedance but the Coulombic efficiency became slightly lower in case of graphite/LCO full cells. When cycling in graphite/NMC full cells, the cell impedance and Coulombic efficiency remained unchanged whereas the capacity retention at elevated temperature was slightly improved. It could be concluded that the impact of TMOBX as an additive strongly depends on the used electrode and electrolyte formulation.

To improve the cycling stability, additives based on an organic heterocyclic ring structure were electrochemically studied. Nie et al. investigated pyridine boron trifluoride type molecules such as pyridine boron trifluoride (PBF) as additives for



Scheme 11 Molecular structures of the film-forming additives

high-voltage applications (Scheme 11) [264]. The structure of PBF consists of two major functional groups, BF_3 as a Lewis acid and an amine Lewis base. Amine functional groups are known as corrosion inhibitors for metals and were therefore considered to decrease the transition metal dissolution from NMC electrode [265]. The BF_3 group as an anion receptor was speculated to react with LiF and dissolve it out of the surface layer resulting in lower impedance [266]. For this reason, PBF type molecules were assumed to be able to control the impedance growth during cycling [131, 254]. The blended electrolyte formulations containing 1 wt% PBF-type molecules together with MMDS or DTD (1 wt%) were investigated in graphite/NMC full cells containing 1 M LiPF_6 EC/EMC (3:7 by wt.) electrolyte (Scheme 11). In presence of MMDS and the PBF-type additives as blend, the impedance of the cells could be reduced. Furthermore, the PBF-type additives helped to increase the Coulombic efficiency of the investigated blends. The PBF + MMDS and PBF + DTD additive blends seemed to be a promising candidate for high-voltage application. However, the electrolyte oxidation could not be suppressed [264]. In addition, by introducing another nitrogen atom into the ring structure a second Lewis acid such as BF_3 is necessary, as studied by Nie et al. [264] and Ma et al. [252]. The authors combined 0.5 wt% pyrazine di-boron trifluoride (PRZ) with 1 wt% MMDS or DTD in 1 M LiPF_6 EC/EMC (3:7 by wt.) in graphite/NMC full cells (Scheme 11). The cycling performance was found to be improved and PRZ + MMDS additive blend was found to be a very effective for LIBs regarding Coulombic efficiency, impedance control and capacity loss. Recently, Nie

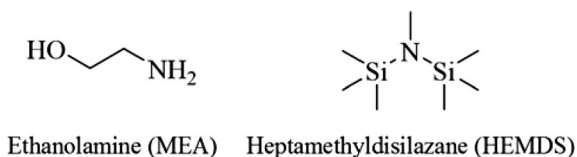
et al. [265] investigated additives consisting of the same amine Lewis base group as PBF but they varied the Lewis acid group. Pyridine phosphorus pentafluoride (PPF) containing a PF_5 instead of BF_3 and pyridine sulfur trioxide (PSO) using a SO_3 as a Lewis acid (Scheme 11). The 1 M LiPF_6 in EC/EMC (3:7 by wt.) electrolyte formulation was enriched either with a single additive or in combination with DTD, VC or triallyl phosphate (TAP) (Scheme 11). TAP was already studied as a single additive for high voltage lithium-ion cells and it was shown that TAP leads to higher Coulombic efficiency since it polymerizes at the surface of the anode and the cathode creating thick stable surface films. However, large amounts of TAP (>3 wt%) led to increased cell impedance [267]. The obtained results indicated high impedance in the cells containing the combination of TAP with PBF or PPF in investigated electrolyte. When comparing the open circuit potential (OCP) after storage at 4.5 V and 60 °C for 500 h, the blend of 2 wt% VC with 1 wt% PSO showed improved storage properties compared to electrolyte containing VC alone. The combination of 2 wt% VC and 1 wt% PBF or PPF did not show any improvement. However, the investigated DTD + PBF and DTD + PFB additive blend-containing electrolytes displayed a significant improvement as confirmed by the higher Coulombic efficiency, lower charge end point capacity slippage, low and stable impedance and no gassing during tests at 4.5 V and 40 °C. In summary aforementioned combination showed high potential for further investigations in high voltage graphite/NMC full cells [265].

4.3 Blended Electrolytes with Salt Stabilizer Additives

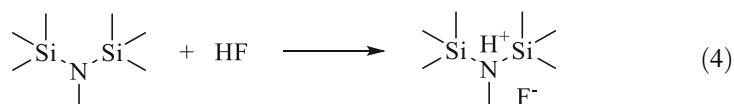
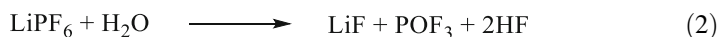
Residual trace amounts of impurities such as water and HF cannot be completely prevented from nonaqueous electrolytes, as well as other LIB cell components even for battery grade products. The formed HF is known as the main reason for the dissolution of cathode materials [67, 131]. For instance, LMO suffers from severe capacity fading after storage at elevated temperature, associated to Mn^{2+} dissolution induced by HF [268, 269]. One solution is based on decreasing the water and HF content by stabilizing the electrolyte. H_2O and HF can be stabilized through hydrogen bonding bridge with molecules such as ethanolamine (MEA) leading to an improved cycling performance [270]. Yamane et al. [271] and Li et al. [272] used single additives hexamethyldisilazane (HMDS) and heptamethyldisilazane (HEMDS) in LiPF_6 -based electrolytes to successfully eliminate HF from the electrolyte. The binary blend of both mentioned additives, MEA + HEMDS, as an electrolyte salt stabilizer was studied by Wu et al. (Scheme 12) [273].

The same group investigated the storage behavior of LMO exposed in air as well as in LMO/Li half-cells at a temperature of 60 °C. The blend of 1 vol% MEA and

Scheme 12 Molecular structures of the salt stabilizer additives



3 vol% HEMDS in 1 M LiPF₆ EC/DMC/EMS (1:1:1) electrolyte was found to suppress Mn²⁺ dissolution as well as the formation of amorphous LMO/electrolyte degradation products, i.e., as λ-MnO₂ with small amounts of MnF₂ and LiF, thus resulting in improved storage performance. The authors proposed the following mechanism: (1) LiPF₆ reacts with water and produces HF (2) and (2) the mentioned degradation products will be generated through a disproportionation reaction of Mn³⁺ at the surface of LiMn₂O₄ induced by HF (3). Theoretical calculations performed by Zhang et al. [274] showed that both H₂O and HF can be stabilized by MEA through hydrogen bridge bonding. HEMDS can eliminate HF by forming the salt compound (4). Since both additives were able to interact with the formed HF simultaneously in different ways, HF can be effectively removed.



4.4 Summary

Using selected additive blends can be more effective than using a single additive, as indicated through the described and discussed examples. This approach represents a very promising way to optimize electrolyte functionalities as selected and investigated additives show ability to compensate for each other's disadvantages. All, in this chapter, reviewed additive blends in corresponding electrolyte formulations are listed in Table 12.

Although, many research groups have been and are still working in this field, additive blended electrolyte formulations still require further and deeper investigation. With this in line, the synergistic effect of the used additives has to be studied in more detail in order to completely understand their main operation, failure mechanisms as well as the synergistic interplay in the electrolyte matrix.

5 Concluding Remarks and Perspectives

One of the critical barriers hindering the further progress and development of LIBs for custom tailored applications is addressed to the electrolyte. In this respect, chemistry and synergy between different electrolyte components are playing a significant role. This review paper highlighted activities in the field of nonaqueous electrolytes containing solvent, salt, as well as additive blends towards improved overall lithium ion cell performance. Blending of properly selected electrolyte constituents appears to be a very promising way to optimize electrolyte functions

Table 12 Overview of the investigated additive blends including the used electrolyte formulation, electrode system as well as their function

Additive blend	Electrolyte	Electrodes	Target improvement	References
TPP + VEC	1.1 M LiPF ₆ EC/EMC (4/6 v/v)	Graphite/ LCO	Flame retardant, SEI	[144, 145]
TPP + BP				
TPP + VA				
TPP + VC				
TCEP + CHB	1 M LiPF ₆ EC/DMC/ EMC (1:1:1 by wt.)	Graphite/ LCO	TCEP: reduced flammability	[154]
			CHB: overcharge protection	
TEP + EC	0.4 M LiTFSI/ PP13-TFSI	Graphite/ LMO	EC: Lower viscosity and increased conductivity	[162]
			TEP: Reduced flammability	
BP + CHB	1 M LiPF ₆ EC/EMC (1:2 v/v)	Graphite/ LCO	Overcharge protection	[180]
VC + PS	BMP-TFSI	LTO/LCO	SEI	[185]
FEC + VC	1 M LiPF ₆ EC/DMC (1:1 by wt.)	Si/Li	SEI	[202–204]
	1 M LiPF ₆ EC/DEC (1:1 by wt.)		Improved cycling performance	
	0.6 M LiTDI EC/DMC (2:1 v/v)			
VC + FA	1 M LiPF ₆ PC	Graphite/Li	SEI	[220, 221]
VC + PS	1 M LiPF ₆ EC/EMC (3:7 v/v)	Graphite/ NMC	SEI	[225, 226]
	1 M LiPF ₆ EC/EMC (3:7 by wt.)	Graphite/Li	SEI	
VC + DTD	1 M LiPF ₆ EC/EMC (3:7 by wt.)	Graphite/ NMC	SEI and CEI	[219, 227]
VC + TMS				
VC + PLS				
VC + PS				
VC + MMDS				
VC + TTSP	1 M LiPF ₆ EC/EMC (3:7 by wt.)	Graphite/ NMC	Improved cycling performance	[236]
VC + TTSPi				
VC + TMS	1 M LiPF ₆ EC/EMC (3:7 by wt.)	Graphite/ NMC	SEI	[250, 251]
VC + DTD			Improved cycling performance	
MMDS + TTSP				
MMDS + TTSPi				

Table 12 continued

Additive blend	Electrolyte	Electrodes	Target improvement	References
PES + MMDS + TTSPi	1 M LiPF ₆ EC/EMC (3:7 by wt.)	Graphite/ NMC442	SEI Improved cycling performance	[250, 252, 254, 255]
VC + MMDS + TTSPi	1 M LiPF ₆ EC/EMC	Graphite/ NMC111	PES: improved cycling performance	[258]
PES + MMDS + TTSPi	(3:7 by wt.)	(4.4 V), Graphite/ NMC442 (4.7 V)	compared to VC	
VC + TMOBX	1 M LiPF ₆ EC/EMC (3:7 by wt.)	Graphite/ LCO Graphite/ NMC	Improved cycling performance	[259, 263]
PBF-type + MMDS	1 M LiPF ₆ EC/EMC (3:7 by wt.)	Graphite/ NMC	Improved cycling performance	[252, 264]
PBF-type + DTD				
PRZ + MMDS				
PRZ + DTD				
PBF + DTD	1 M LiPF ₆ EC/EMC (3:7 by wt.)	Graphite/ NMC	TAP + PBF, TAP + PPF: High impedance	
PBF + VC				
PBF + TAP				
PPF + DTD				
PPF + VC				
PPF + TAP				
PSO + DTD				
PSO + VC				
PSO + TAP				
VC + PBF, VC + PPF:	[265]			
No improvement				
DTD + PBF,				
DTD + PPF:				
Improved cycling performance				
MEA + HEMDS	1 M LiPF ₆ EC/DMC/ EMS (1:1:1 by wt.)	LMO/Li	Decreased Mn dissolution Stabilization of HF and water	[273, 274]

through the ability of blended components to compensate for each other's disadvantages or to eliminate undesired disadvantages. With this in line, the effect of different solvent, salt as well as additive blends on the physicochemical and electrochemical features of nonaqueous electrolyte formulations was discussed using interesting literature examples, pointing out the influence on anode, cathode, suppression or inhibition of Al dissolution, thermal stability as well as the overall cell performance.

However, in respect to the synergistic effect of afore described and discussed electrolyte blends, there are still many questions left unanswered, thus acting as a barrier towards desired improvements and refined electrolyte optimization. In order to systematically develop novel blended nonaqueous electrolyte formulations, the following points must be taken in consideration and studied in more detail: correlation between the molecular and electronic structure of the electrolyte components and their reactivity, the synergistic effects of the blended components, main operation and failure processes, study on intermediate species, investigation of the different plausible reaction pathways as well as the analysis of the limiting and determining steps which provide rationalization of the obtained results. Integrating the knowledge gained from profound studies of the aforementioned parameters and processes will lead to desired improvements that positively affect lithium ion cell chemistry and LIB performance.

We hope that this review provides a good overview and clear perspective on blended nonaqueous electrolyte formulations thus raising the motivation in encouraging researchers to proceed with further and deeper investigations in this promising field.

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